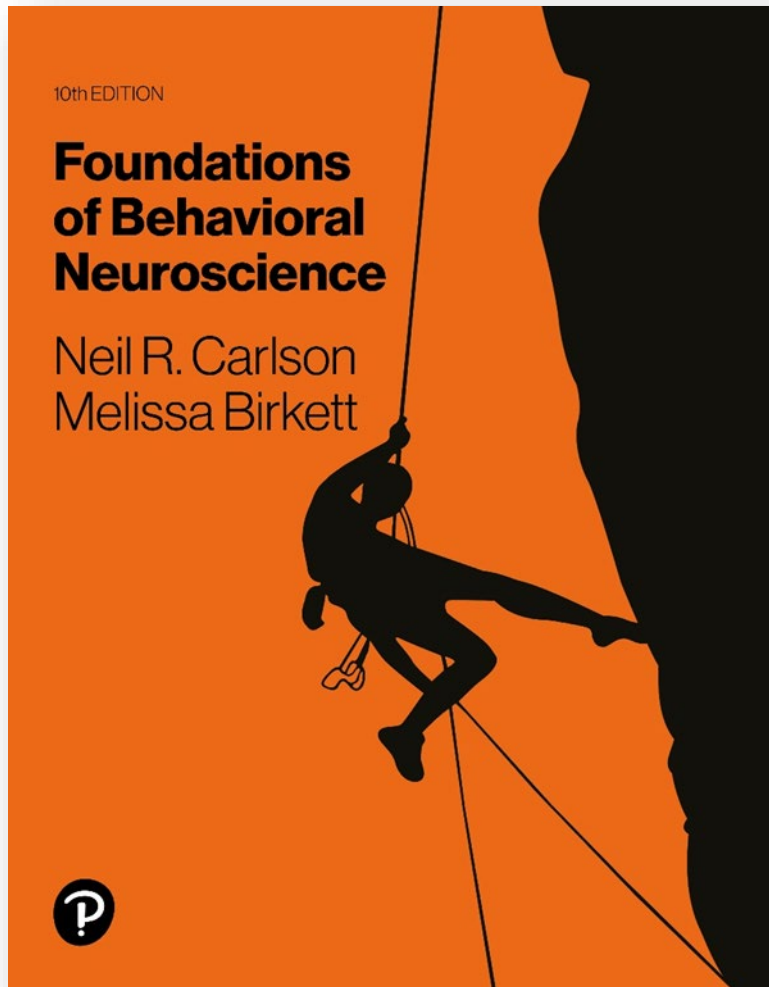


Foundations of Behavioral Neuroscience

Tenth Edition



Chapter 17

Neurodevelopmental Disorders, Stress, Posttraumatic Stress Disorder, and Substance Use Disorders

Neurodevelopmental Disorders

- **Autism Spectrum Disorder (ASD)**
- **Attention-Deficit/Hyperactivity Disorder**

Autism Spectrum Disorder (ASD) (1 of 5)

Learning Objective 17.1 Describe the symptoms and possible causes of autism spectrum disorder.

- Failure to develop typical social relations
- Impaired development of communicative ability
- Presence of repetitive, stereotyped behaviors, fixated interests, or inflexible adherence to routines

Autism Spectrum Disorder (ASD) (2 of 5)

Symptoms

- Four times more common in males
- Present from early childhood and limits or impairs everyday functioning
- One in 31 (3.2%) 8-year-old children have been identified with autism spectrum disorder (ASD), according to an analysis published today in CDC's *Morbidity and Mortality Weekly Report (MMWR)*.

Early report

Ileal-lymphoid-nodular hyperplasia, non-specific colitis, and pervasive developmental disorder in children

A J Wakefield, S H Murch, A Anthony, J Linnell, D M Casson, M Malik, M Berelowitz, A P Dhillon, M A Thomson, P Harvey, A Valentine, S E Davies, J A Walker-Smith

Summary

Background We investigated a consecutive series of children with chronic enterocolitis and regressive developmental disorder.

Methods 12 children (mean age 6 years [range 3–10], 11 boys) were referred to a paediatric gastroenterology unit with a history of normal development followed by loss of acquired skills, including language, together with diarrhoea and abdominal pain. Children underwent gastroenterological, neurological, and developmental assessment and review of developmental records. Ileocolonoscopy and biopsy sampling, magnetic-resonance imaging (MRI), electroencephalography (EEG), and lumbar puncture were done under sedation. Barium follow-through radiography was done where possible. Biochemical, haematological, and immunological profiles were examined.

Findings Onset of behavioural symptoms was associated by the parents, with measles, mumps, and rubella vaccination in eight of the 12 children, with measles infection in one child, and otitis media in another. All 12 children had intestinal abnormalities ranging from lymphoid nodular hyperplasia to atypical ulceration. Histology showed patchy chronic inflammation in seven, in 11 children and reactive ileal lymphoid hyperplasia in seven, but no granulomas. Behavioural disorders included autism (nine), disintegrative psychosis (one), and possible postviral or vaccinal encephalitis (two). There were no focal neurological abnormalities and MRI and EEG tests were normal. Abnormal laboratory results were significantly raised urinary methylmalonic acid compared with age-matched controls ($p=0.03$), low haemoglobin in four children, and low serum IgA in four children.

Interpretation We identified associated gastrointestinal disease and developmental regression in a group of previously normal children, which was generally associated in time with possible environmental triggers.

Lancet 1998; **351**: 637–41

See Commentary page

Inflammatory Bowel Disease Study Group, University Departments of Medicine and Histopathology (A J Wakefield FRCS, A Anthony MR, J Linnell PhD, A P Dhillon MRCPH, S E Davies MRCPH) and the **University Departments of Paediatric Gastroenterology** (S H Murch MB, D M Casson MRCP, M Malik MRCP, M A Thomson FRCP, J A Walker-Smith FRCP), **Child and Adolescent Psychiatry** (M Berelowitz FRCPsych), **Neurology** (P Harvey FRCP), and **Radiology** (A Valentine FRCS), **Royal Free Hospital and School of Medicine, London NW3 2QG, UK**

Correspondence to: Dr A J Wakefield

Introduction

We saw several children who, after a period of apparent normality, lost acquired skills, including communication. They all had gastrointestinal symptoms, including abdominal pain, diarrhoea, and vomiting and, in some cases, food intolerance. We describe the clinical findings, and gastrointestinal features of these children.

Patients and methods

12 children, consecutively referred to the department of paediatric gastroenterology with a history of a pervasive developmental disorder with loss of acquired skills and intestinal symptoms (diarrhoea, abdominal pain, bloating and food intolerance), were investigated. All children were admitted to the ward for one week, accompanied by their parents.

Clinical investigations

We took histories including details of immunisations and exposure to infectious diseases, and assessed the children. In 11 cases the history was obtained by the senior clinician (JW-S). Neurological and psychiatric assessments were done by consultant staff (PH, MB) with HMS-4 criteria.¹ Developmental assessments included a review of prospective developmental records from parents, health visitors, and general practitioners. Four children did not undergo psychiatric assessment in hospital; all had been assessed professionally elsewhere, so these assessments were used as the basis for their behavioural diagnosis.

After bowel preparation, ileocolonoscopy was performed by SHM or MAT under sedation with midazolam and pethidine. Paired frozen and formalin-fixed mucosal biopsy samples were taken from the terminal ileum; ascending, transverse, descending, and sigmoid colons, and from the rectum. The procedure was recorded by video or still images, and were compared with images of the previous seven consecutive paediatric colonoscopies (four normal colonoscopies and three on children with ulcerative colitis), in which the physician reported normal appearances in the terminal ileum. Barium follow-through radiography was possible in some cases.

Also under sedation, cerebral magnetic-resonance imaging (MRI), electroencephalography (EEG) including visual, brain stem auditory, and sensory evoked potentials (where compliance made these possible), and lumbar puncture were done.

Laboratory investigations

Thyroid function, serum long-chain fatty acids, and cerebrospinal-fluid lactate were measured to exclude known causes of childhood neurodegenerative disease. Urinary methylmalonic acid was measured in random urine samples from eight of the 12 children and 14 age-matched and sex-matched normal controls, by a modification of a technique described previously.² Chromatograms were scanned digitally on computer, to analyse the methylmalonic-acid zones from cases and controls. Urinary methylmalonic-acid concentrations in patients and controls were compared by a two-sample *t* test. Urinary creatinine was estimated by routine spectrophotometric assay.

Children were screened for antidiomycal antibodies and boys were screened for fragile-X if this had not been done

a 1998 study that fraudulently claimed a link between the measles, mumps, and rubella (MMR) vaccine and autism.

Disgraced anti-vaxxer Andrew Wakefield aims to advance his agenda in Texas election

Wakefield sees Houston primary as a new start after his anti-vaccine theories were debunked and medical license revoked



▲ Andrew Wakefield prompted large drops in vaccination rates in the UK and Ireland when he published a now-retracted paper in 1998. Photograph: Anthony Devlin/PA

Anti-vaccine campaigners have found a growing political voice for their debunked

Advertisement



New York

EST

month old

Books on Andrew Wakefield

- *The Doctor Who Fooled the World*
- Paul Offitt's *Autism's False Prophets* (2008)
- Seth Mnookin's *The Panic Virus* (2012)

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🏠 — [Global Center for Health Security](#) — [The Transmission](#) — Now disgraced doctor Andrew Wakefield targets mumps vaccine with new Hollywood feature film which claims 'dangerous' jab causes serious long-term health issues

Now disgraced doctor Andrew Wakefield targets mumps vaccine with new Hollywood feature film which claims 'dangerous' jab causes serious long-term health issues

PUBLISHED Feb 14, 2024

MSN

Disgraced doctor Andrew Wakefield – whose fake anti-vax data has been blamed for the current measles epidemic – has set his sights on discrediting the mumps vaccine.

The 67-year-old, who fled to the US after being struck off in the UK for fraudulently connecting the measles, mumps and rubella (MMR) vaccine to autism, is seeking Hollywood fame with his first feature film Protocol-7, which claims the mumps jab causes serious long-term health issues.

RESOURCE LINKS

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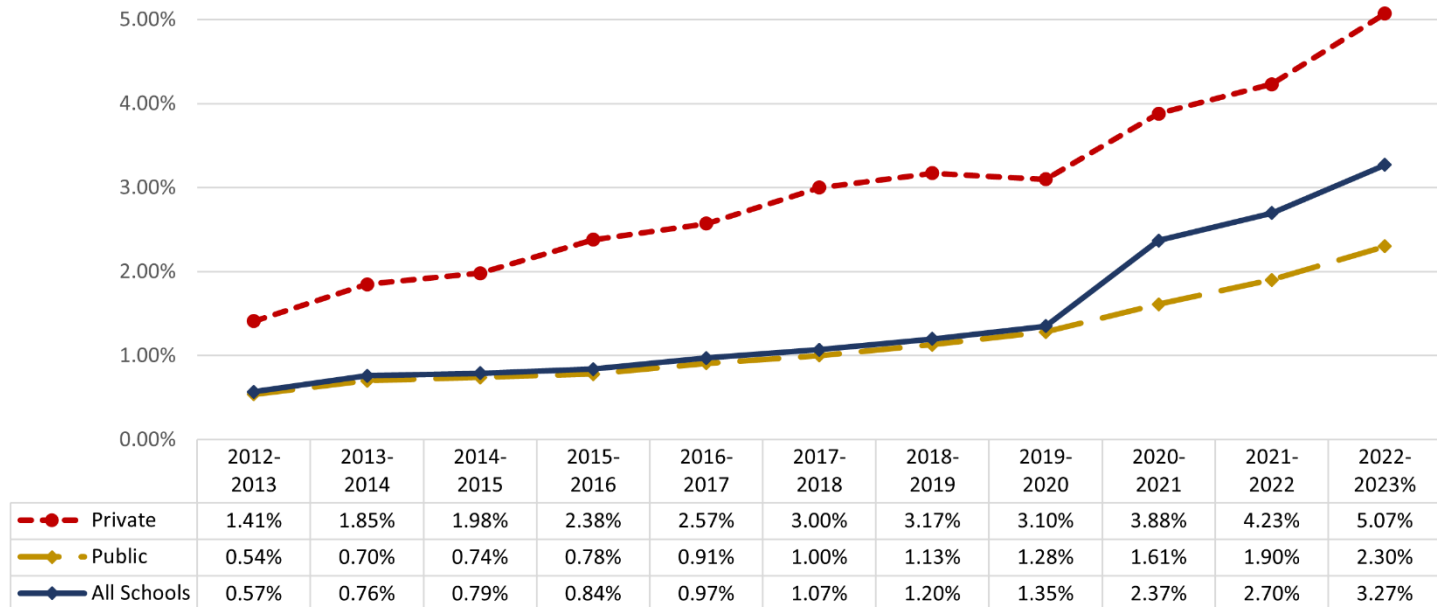
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MMR-vaccine coverage in the United Kingdom fell to around 80% in the mid-2000s (from the necessary 95%), leading to outbreaks.

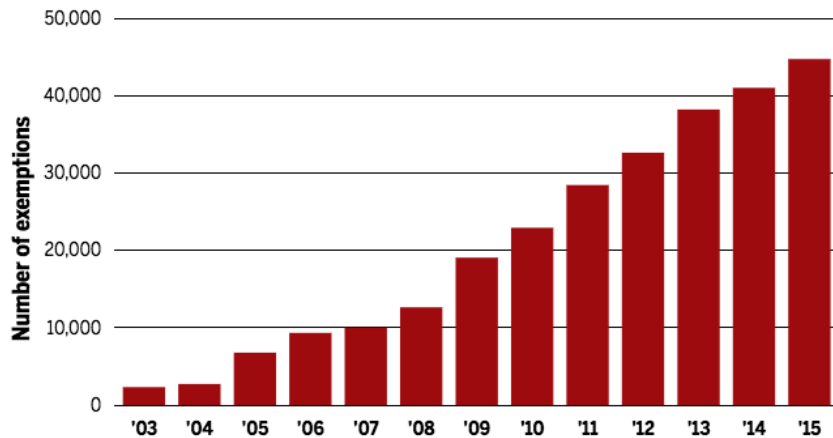
Percent of Students with ≥ 1 Conscientious Exemption on File, by School Type, Texas 2012-2023



Nonmedical exemptions in Texas



Nonmedical exemptions for K–12th grade students, TX



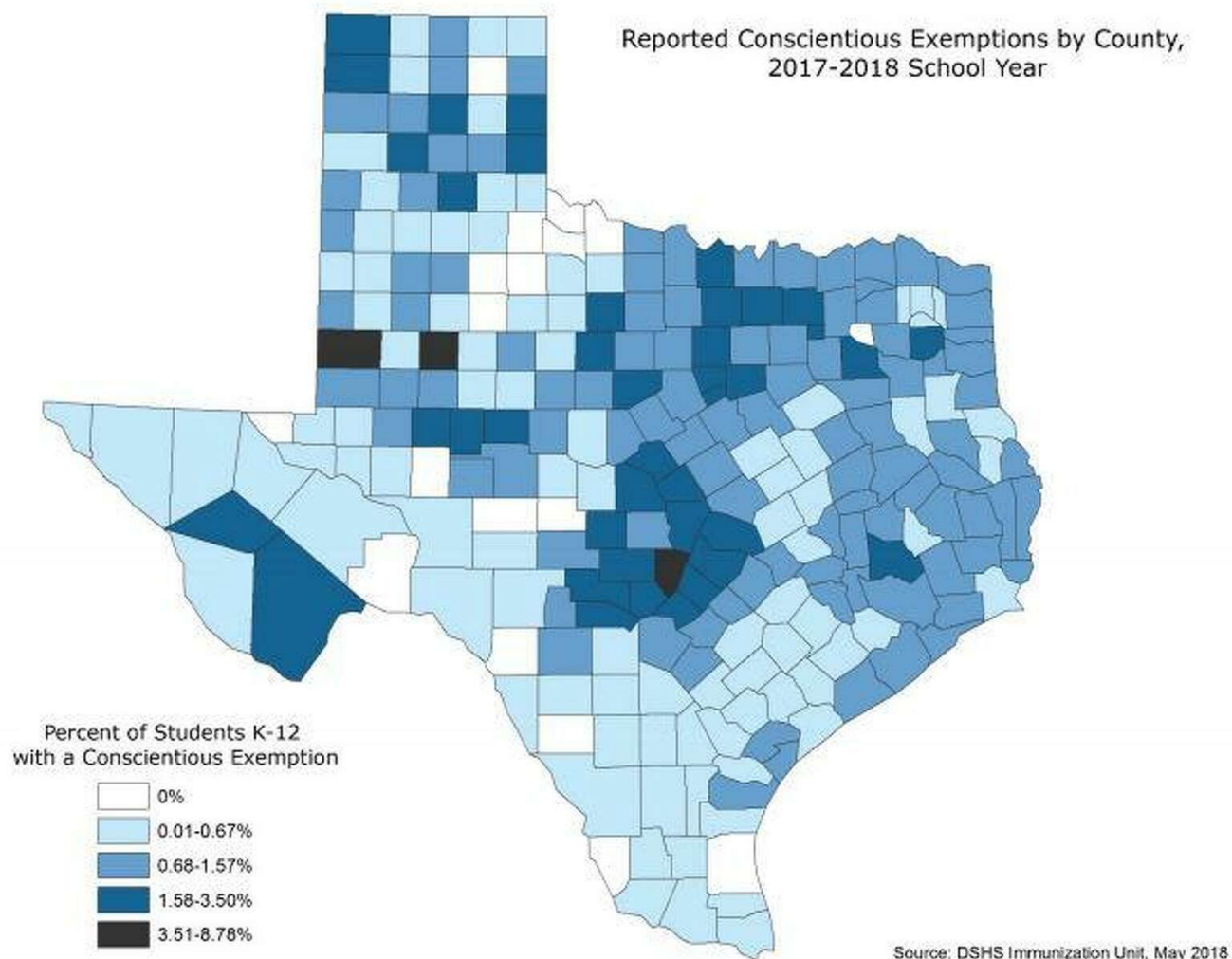
Martin Enserink/Science

Data: Self-Reported by Public ISDs and Accredited Private Schools, Annual Report of Immunization Status, DSHS, Immunization Branch

Measles is an extremely contagious pathogen and often the first one to spread when vaccination rates dip below about 95%. The risk of outbreaks is even greater because unvaccinated children aren't randomly distributed. In Gaines county in western Texas, for instance, the exemption rate is already 4.8%, and at one school in Austin, it's 40%. "I would describe Texas as sitting on a ticking time bomb," Wootton says.

Not everyone is so gloomy. Some counties in Washington and Colorado have higher levels of exemptions, says immunologist Diane Griffin of the Johns Hopkins Bloomberg School of Public Health in Baltimore, Maryland. "I don't think that Texas is any worse than a number of other states, but pointing out the problem and the solution is important," she wrote in an email.

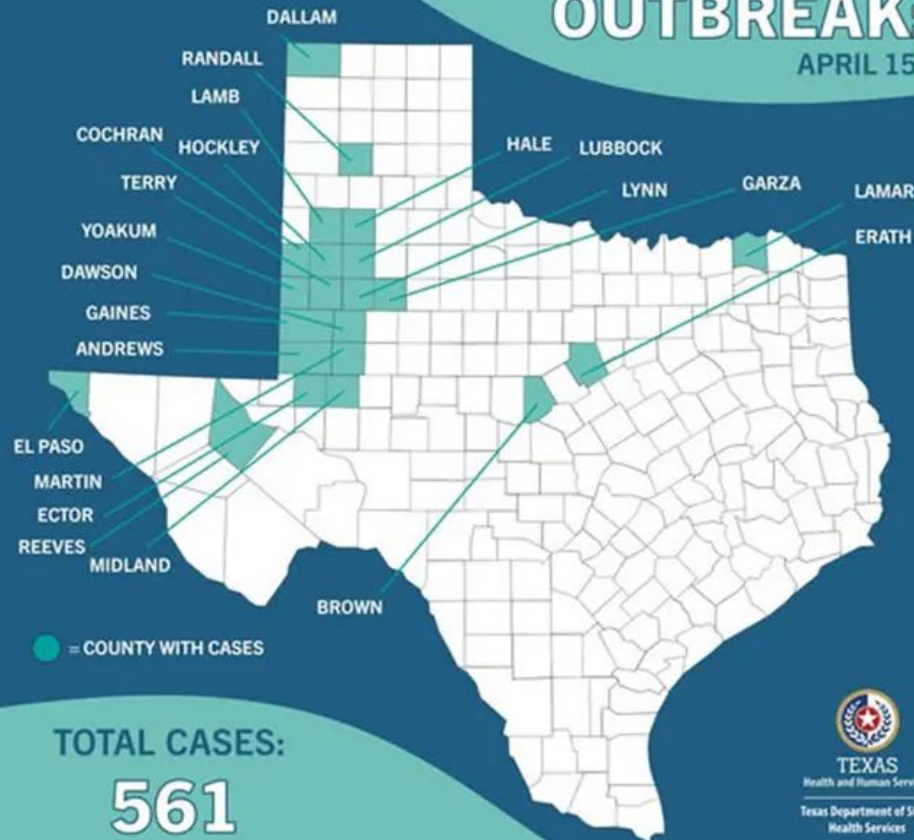
But Hotez believes the situation in the Lone Star State is more perilous. One factor is the arrival of Wakefield, widely seen as the father of the modern antivaccine movement. Wakefield published a paper in *The Lancet* in 1998 that alleged a link between the MMR vaccine (which combines shots against measles, mumps, and rubella) and autism. Several large studies have failed to find the link

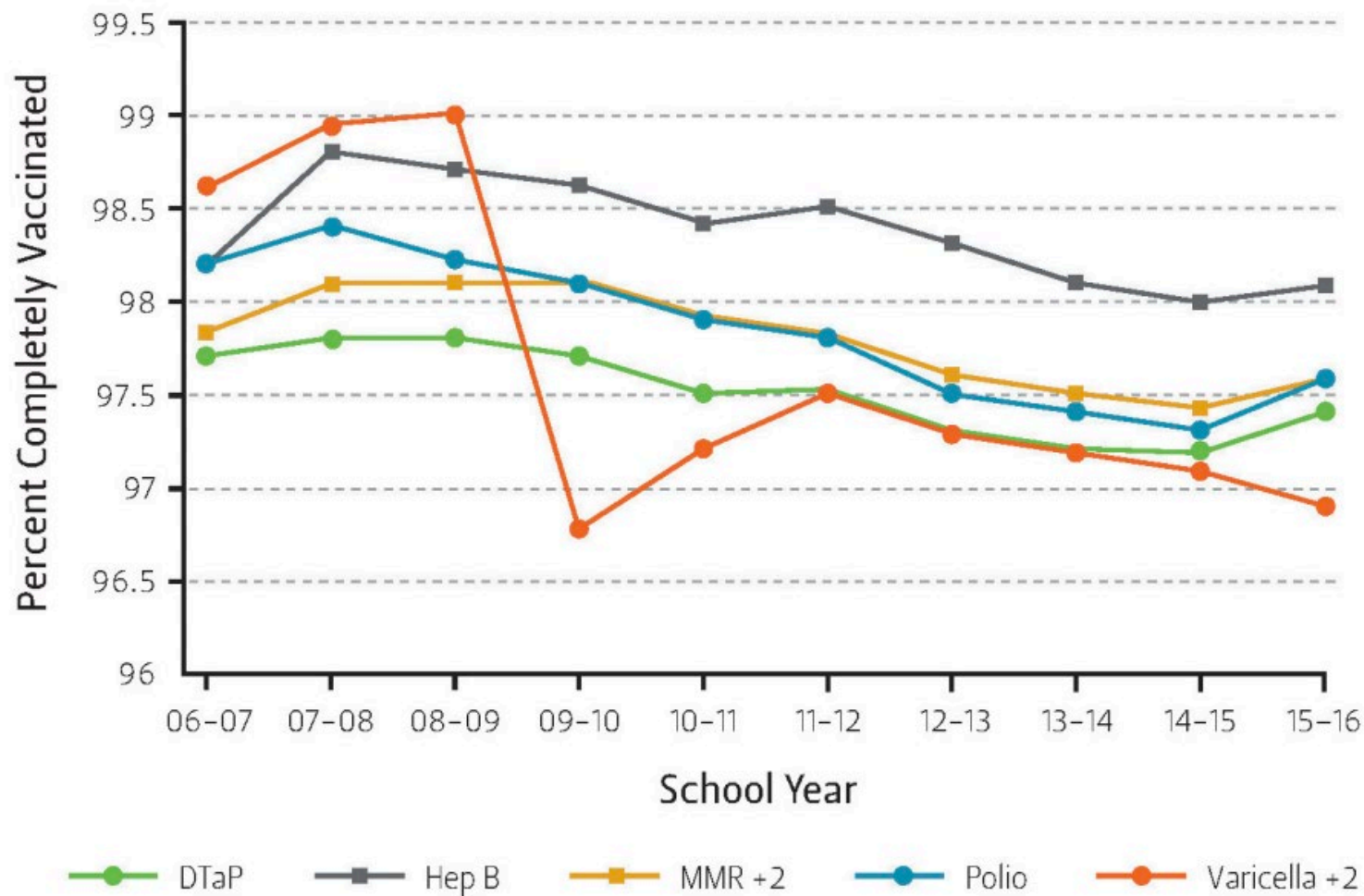


Note: Confirmed cases are counted where the person with measles lives.

MEASLES OUTBREAK:

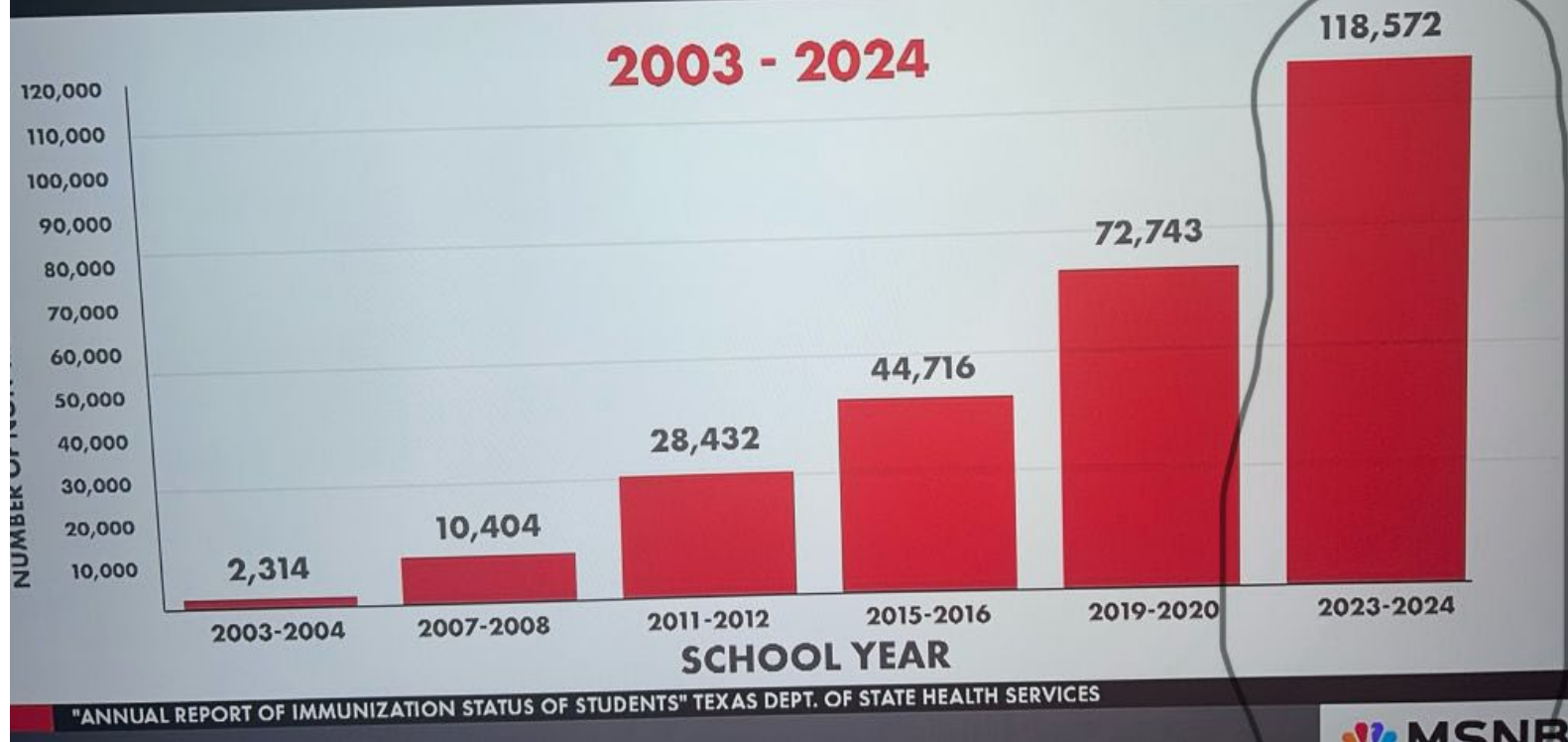
APRIL 15





Source Texas Annual Report of Immunization Status,
Immunization Branch, Texas Department of State Health Services.

TEXAS K-12 PERSONAL BELIEF VACCINE EXEMPTIONS



Roald Dahl's Heartbreaking Letter About Losing his Daughter in 1962

Olivia, my eldest daughter, caught measles when she was seven years old. As the illness took its usual course I can remember reading to her often in bed and not feeling particularly alarmed about it. Then one morning, when she was well on the road to recovery, I was sitting on her bed showing her how to fashion little animals out of coloured pipe-cleaners, and when it came to her turn to make one herself, I noticed that her fingers and her mind were not working together and she couldn't do anything.

"Are you feeling all right?" I asked her.

"I feel all sleepy," she said.

In an hour, she was unconscious. In twelve hours she was dead.



https://www.thedailysheepie.com/california-begins-injecting-children-with-mercury-state-safety-laws-nullified-to-push-vaccines_122015

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TUESDAY, APRIL 16TH, 2019



FEATURED

**SIX STATE E.COLI OUTBREAK
BLAMED ON GROUND BEEF**

**MUSLIM RIOTS LEAVE COPENHAGEN
IN FLAMES**

**TEEN KILLS GRANDPARENTS THEN
CONFESSES TO PLANS TO "CAUSE
HARM" AT HIS HIGH SCHOOL.**

**NOTRE DAME CATHEDRAL IN PARIS
ON FIRE**

**H&R BLOCK: TAXES DOWN 25
PERCENT NATIONWIDE DUE TO
TRUMP'S CUTS**

**CALIFORNIA BEGINS INJECTING CHILDREN WITH
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THE BIRTH

California Health and Human Services Agency Secretary Diana S. Dooley

Thimerosal and Vaccines

- Thimerosal is a mercury-containing organic compound
- Thimerosal, which is approximately 50% mercury by weight, has been one of the most widely used preservatives in vaccines. It is metabolized or degraded to ethylmercury and thiosalicylate. Ethylmercury is an organomercurial that should be distinguished from methylmercury, a related substance that has been the focus of considerable study.
- Methylmercury is the type of mercury found in certain kinds of fish. At high exposure levels, methylmercury can be toxic to people. In the United States, federal guidelines keep as much methylmercury as possible out of the environment and food, but over a lifetime, everyone is exposed to some methylmercury.

Thimerosal and Vaccines

- All vaccines routinely recommended for children 6 years of age and younger in the U.S. are available in formulations that do not contain thimerosal.
- Vaccines that do not contain thimerosal as a preservative are also available for adolescents and adults.
- A robust body of peer-reviewed scientific studies conducted in the U.S. and other countries support the safety of thimerosal-containing vaccines.
- Preservatives prevent microbial growth.
- A preservative is required in multi-dose vials of vaccines.
- The use of thimerosal as a preservative in vaccines has markedly declined due to reformulation and development of new vaccines in single-use presentations.

Autism causes

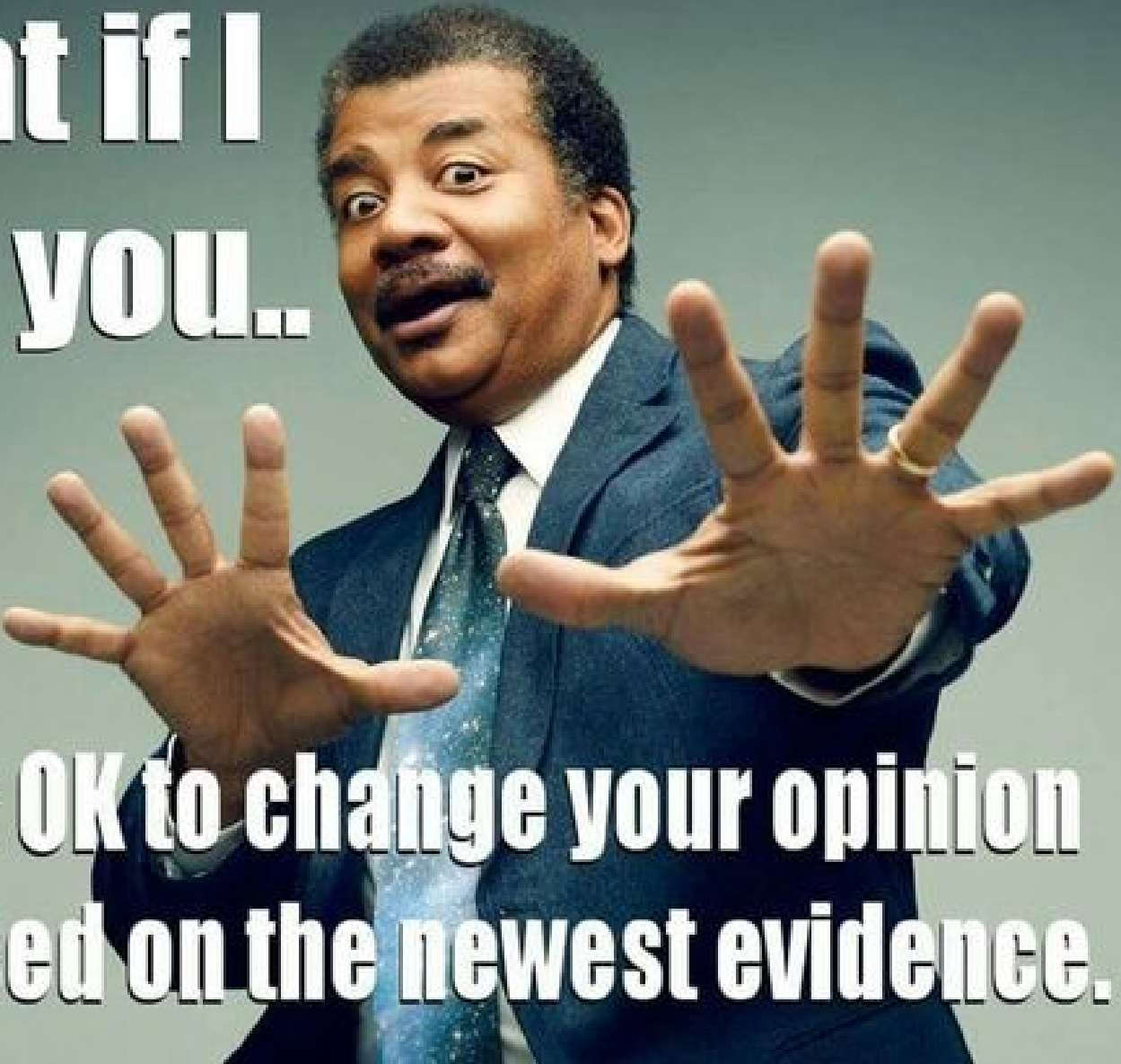
- Scientific Method
 - State an explicit problem, based upon observation and experimentation.
 - Test or criticize a hypothesis through experimentation.
 - Interpret the data and come to a conclusion, ideally using mathematics.
 - Publish the findings

Conspiracy Method

- State an explicit problem, based upon observation and experimentation.
- Make something up that benefits them
 - Usually based on a kernel of truth
- When science does not support their findings then it is because the government is against them
- Do not publish the findings. Go to the people. Do not provide any evidence for their findings
- Never change their mind

Conspiracy Method

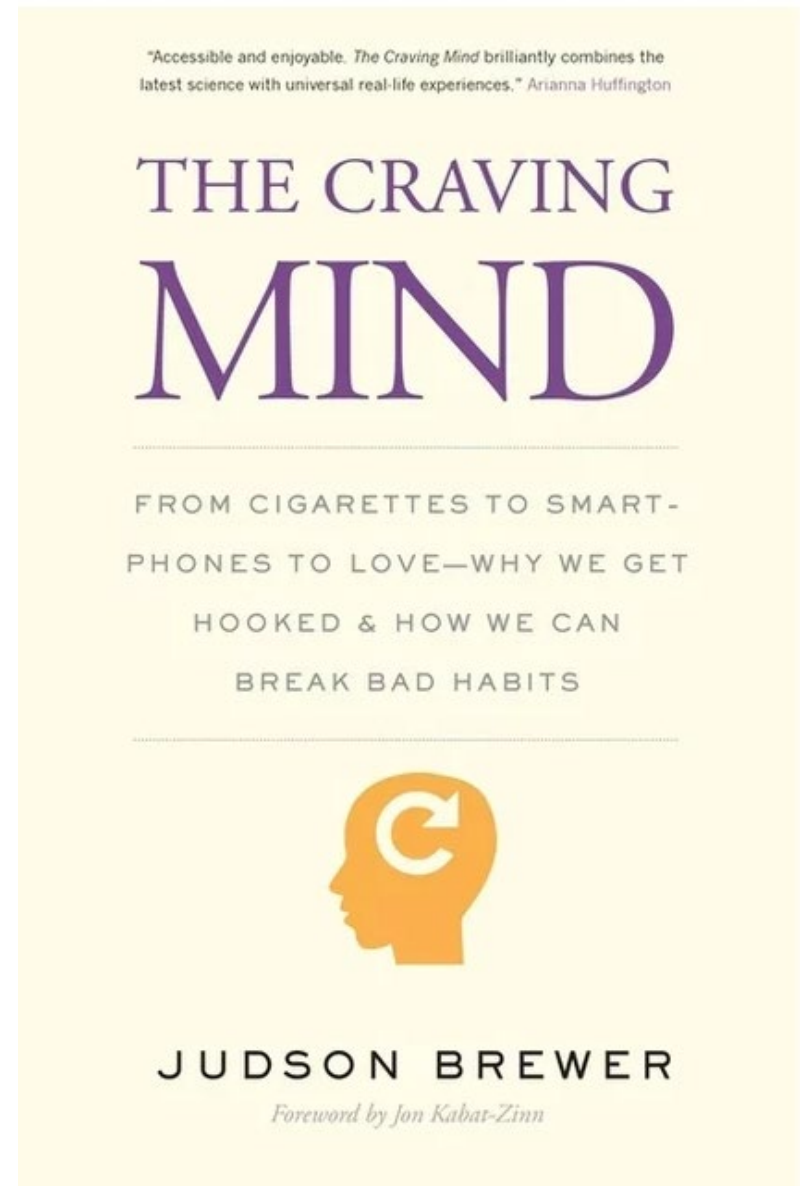
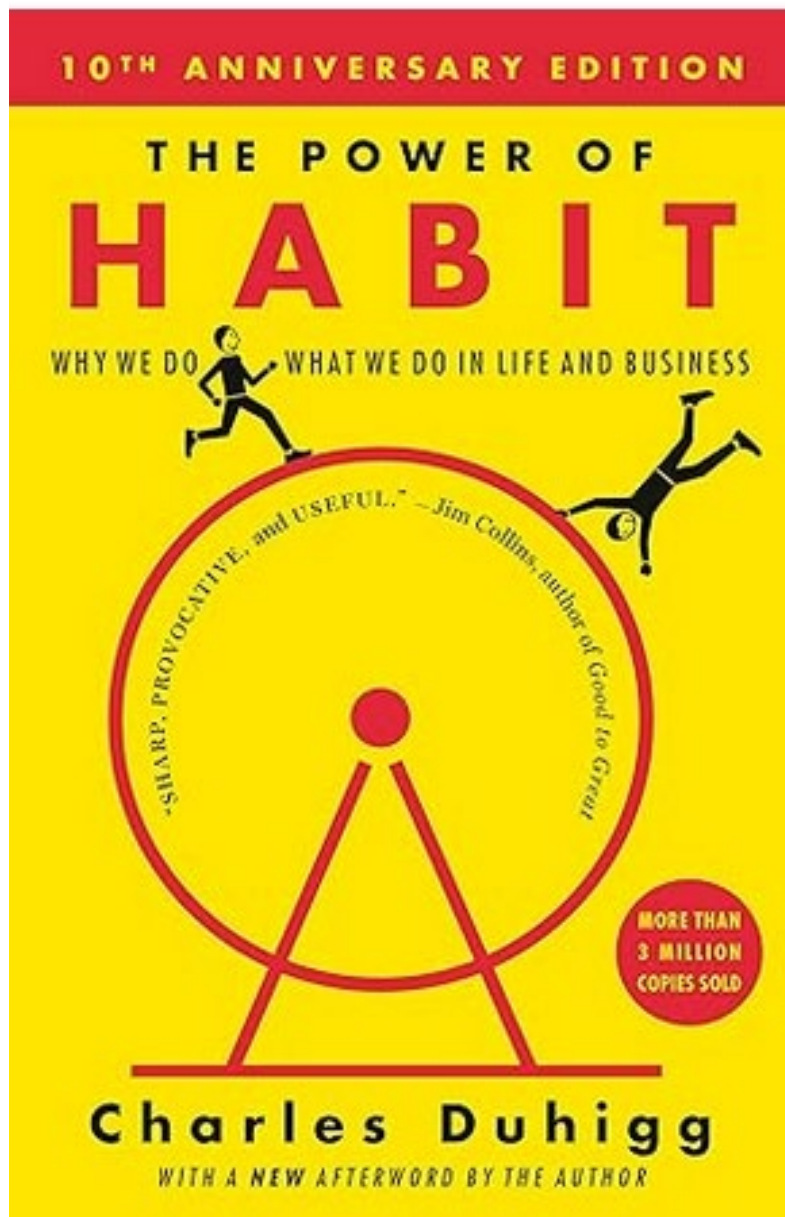
- Can you trace the source?
- Is it possible to refute the argument?

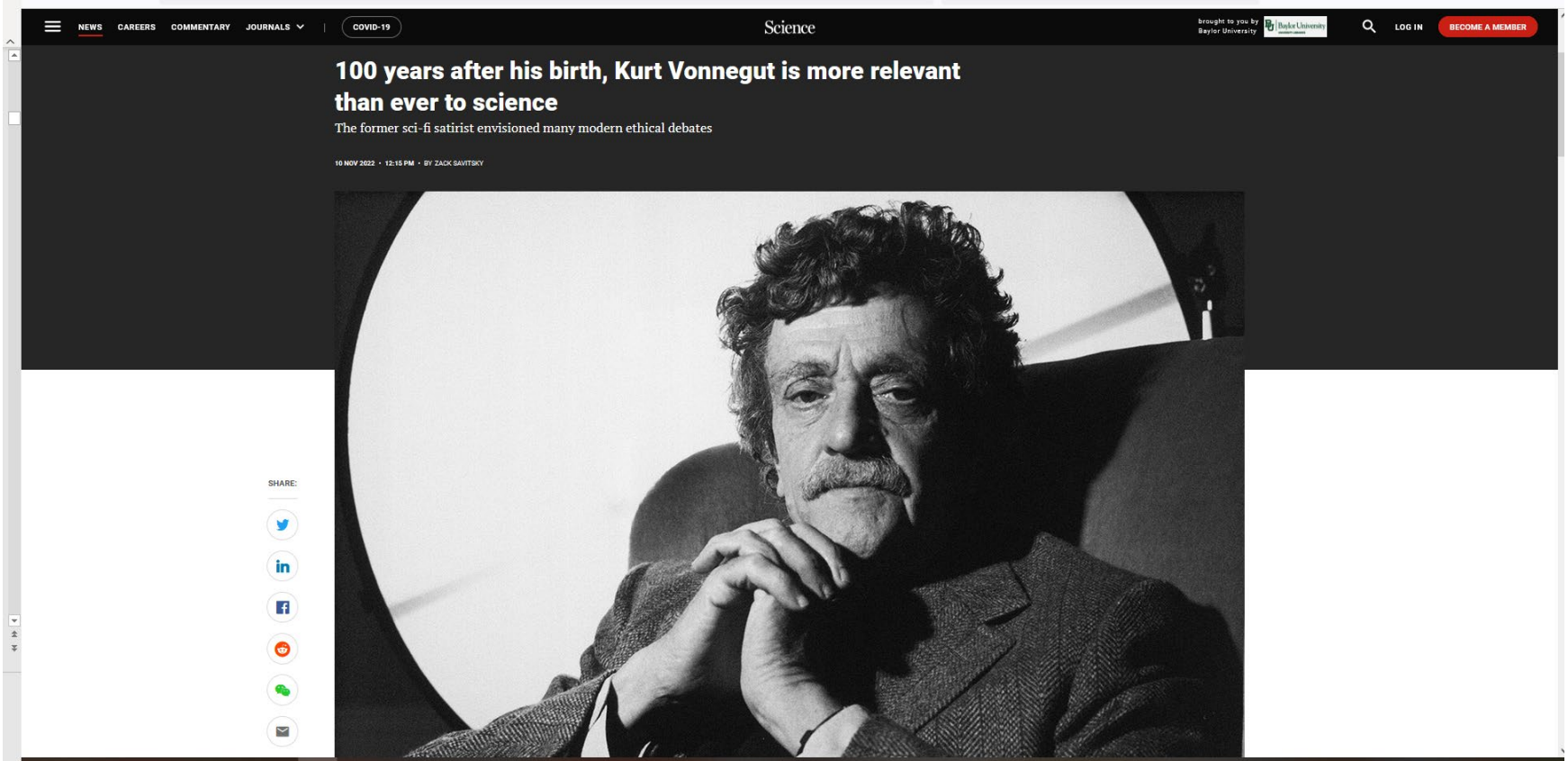
Neil deGrasse Tyson is shown from the chest up, wearing a dark blue suit, a white shirt, and a blue tie with a white star pattern. He has a surprised or emphatic expression on his face, with wide eyes and an open mouth. His hands are held out in front of him, palms facing forward, with fingers spread. The background is a plain, light blue-grey color.

**What if I
told you..**

**It's OK to change your opinion
based on the newest evidence.**







“

Science is magic that works.

CAT'S CRADLE

**“I’m often asked whether
I believe in global warming.
I now just reply with the question:
Do you believe in gravity?”**

—Neil deGrasse Tyson
*Director, Hayden Planetarium,
American Museum of Natural History*

photo: Greyhawk68

Long-term measles-induced immunomodulation increases overall childhood infectious disease mortality

Michael J. Mina,^{1,2*} C. Jessica E. Metcalf,^{1,3} Rik L. de Swart,⁴
A. D. M. E. Osterhaus,⁴ Bryan T. Grenfell^{1,3}

Immunosuppression after measles is known to predispose people to opportunistic infections for a period of several weeks to months. Using population-level data, we show that measles has a more prolonged effect on host resistance, extending over 2 to 3 years. We find that nonmeasles infectious disease mortality in high-income countries is tightly coupled to measles incidence at this lag, in both the pre- and post-vaccine eras. We conclude that long-term immunologic sequelae of measles drive interannual fluctuations in nonmeasles deaths. This is consistent with recent experimental work that attributes the immunosuppressive effects of measles to depletion of B and T lymphocytes. Our data provide an explanation for the long-term benefits of measles vaccination in preventing all-cause infectious disease. By preventing measles-associated immune memory loss, vaccination protects polymicrobial herd immunity.

Autism Spectrum Disorder Causes

- Strongly heritable
- Caused by a wide variety of rare mutations, especially those that interfere with neural development and communication
- Concordance rate for the more broadly defined autistic spectrum disorders (ASD), is 70-90 percent for monozygotic twins and 5-10 percent for dizygotic twins (Sebat et al. 2007)

Autism Spectrum Disorder (ASD) (3 of 5)

Possible Causes

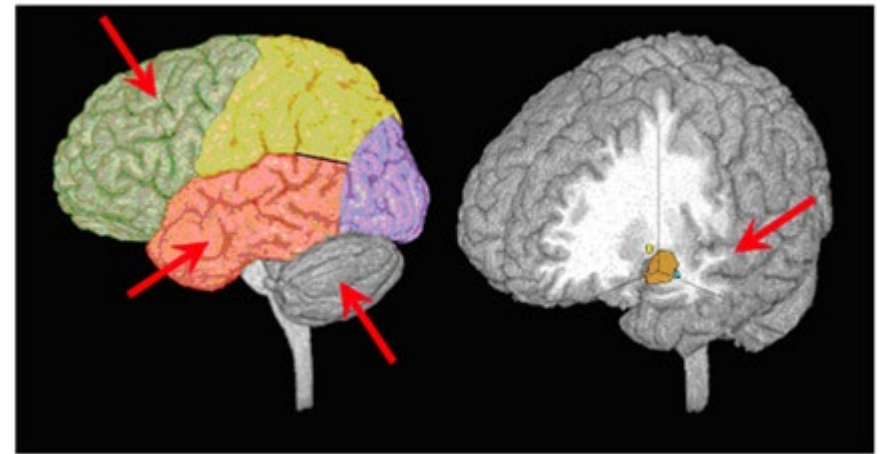
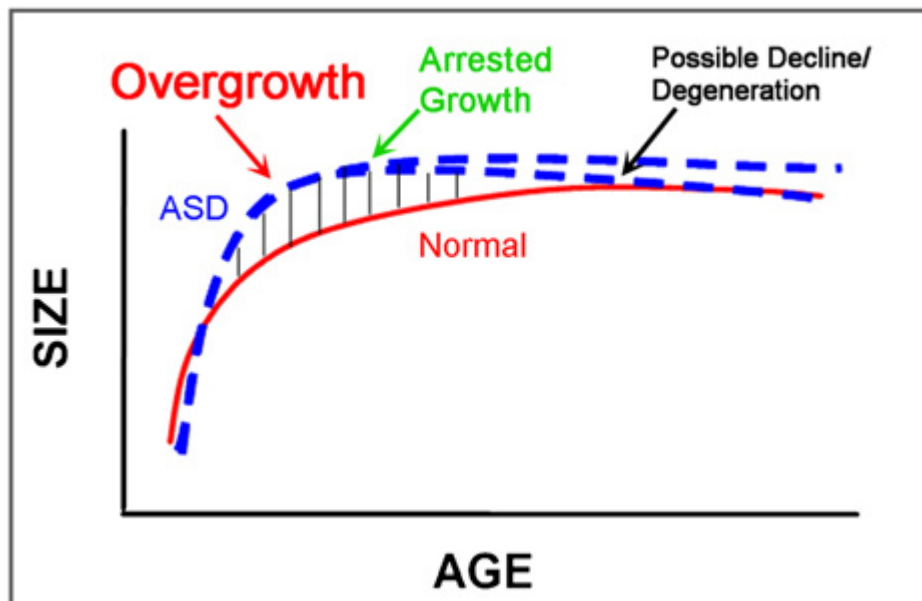
- 20% of children with an older sibling with autism are likely to also have autism



Autism Spectrum Disorder (ASD) (3 of 5)

Possible Causes

- Not all parts of the brain show the same growth
- The development of a large number of short-range axons inhibits the development of long-range axons



Environmental Factors of Interest

Maternal Infection

12% increase in risk of offspring developing ASD (Jiang et al, 2016)

30% increase when requiring hospitalization

Maternal Stress

Third-trimester prenatal stress increased the risk of ASD development in offspring (Class et al, 2014)

Environmental Factors of Interest

Dietary Factors

- Prenatal

In a small cohort, mothers in the lowest 5% of omega-3 fatty acid intake had a significant increase in offspring ASD risk (Lyll et al, 2013)

- Postnatal

Supplementation with DHA in a sample of 30 children with ASD resulting in improvement of the CARS score in 66% of individuals (Meguid et al, 2008)

Autism Spectrum Disorder (ASD) (4 of 5)

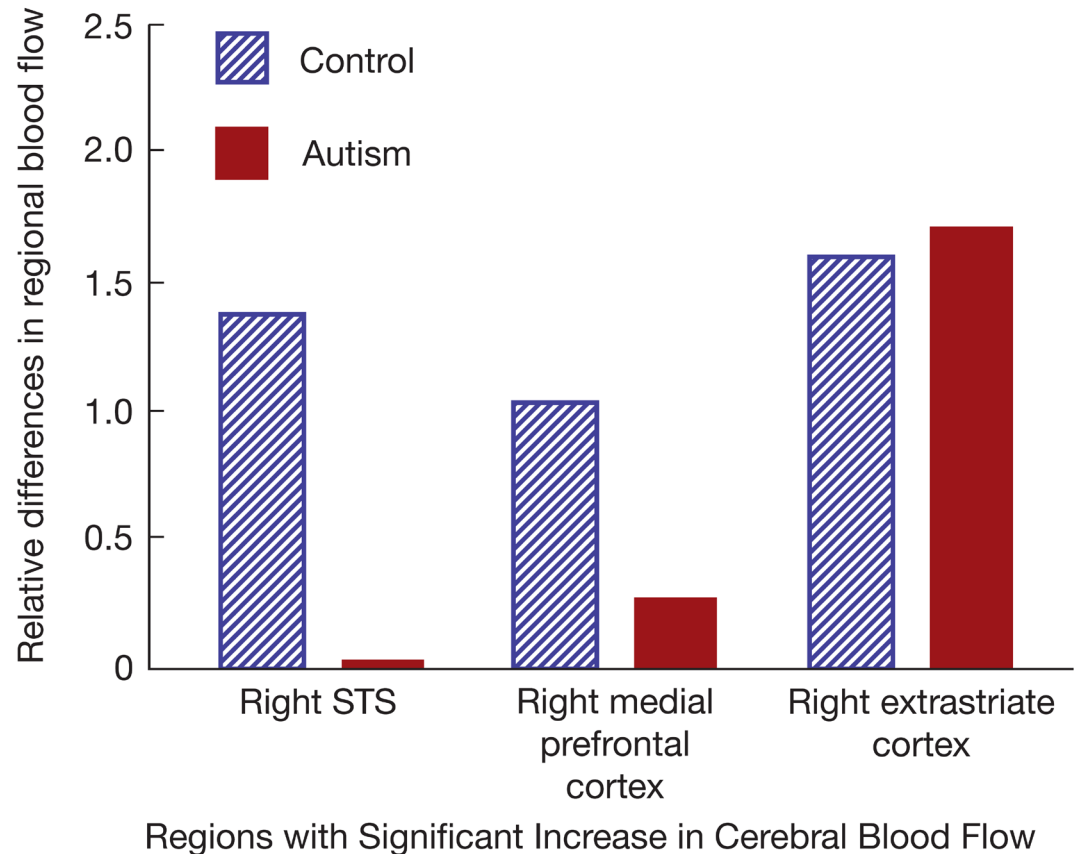
Possible Causes continued

- Changes in brain prevent formation of a theory of mind
 - Difficulty inferring thoughts, feelings, and intentions of others

Figure 17.1

Theory of Mind

The graph shows relative activation of specific brain regions of adults with autism and control participants viewing a “theory of mind” animation of two triangles moving interactively with implied intentions. STS = superior temporal sulcus. Source: Based on Castelli, F., Frith, C., Happé, F., & Frith, U. (2002). Autism, Asperger syndrome and brain mechanisms for the attribution of mental states to animated shapes. *Brain*, 125(8), 1839–1849.





This is Sally



This is Anne



Sally puts her ball in the basket.



Sally goes away.



Anne moves the ball to her box.



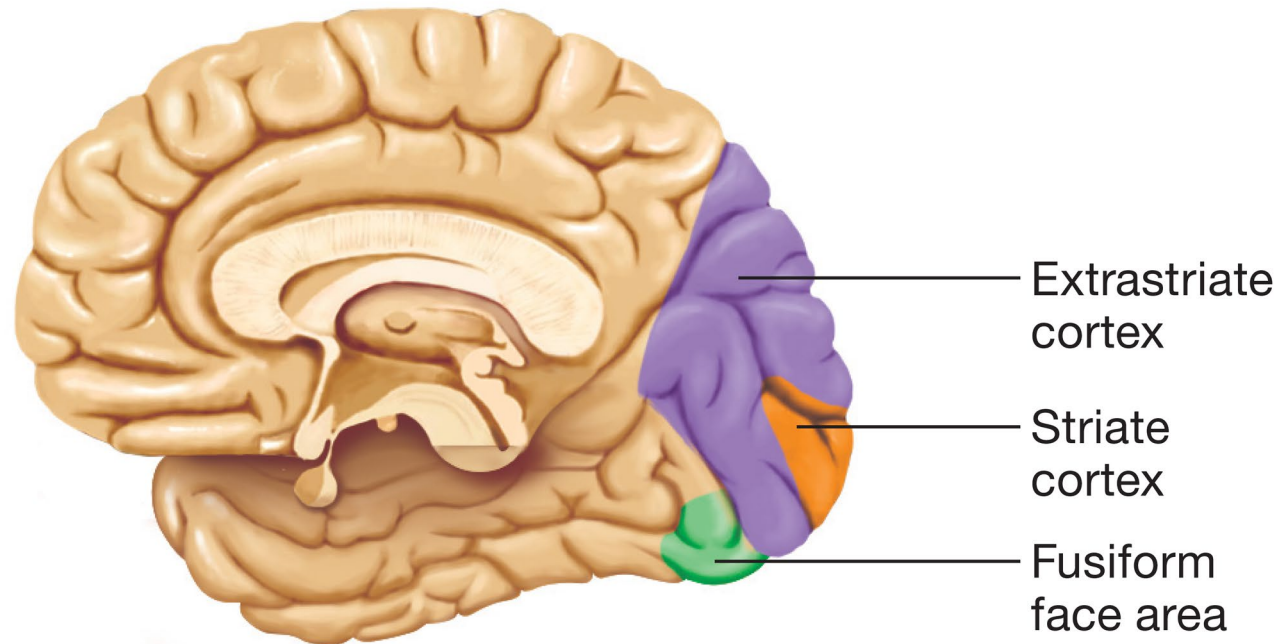
Where will Sally look for her ball?

Autism Spectrum Disorder (ASD) (4 of 5)

- Structural- and functional-imaging research to investigate the neural basis
 - Lower activation of superior temporal sulcus (STS) and medial prefrontal cortex
 - FFA fails to respond to human faces due to lack of interest
 - Abnormal development of the mirror neuron system

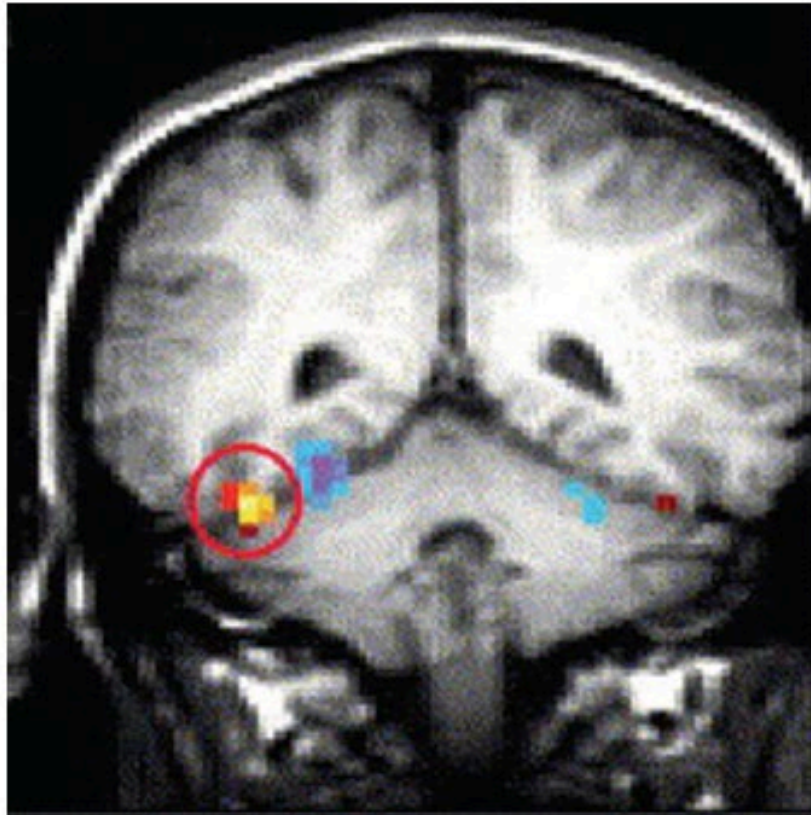
Figure 17.2

Fusiform Face Area and Autism

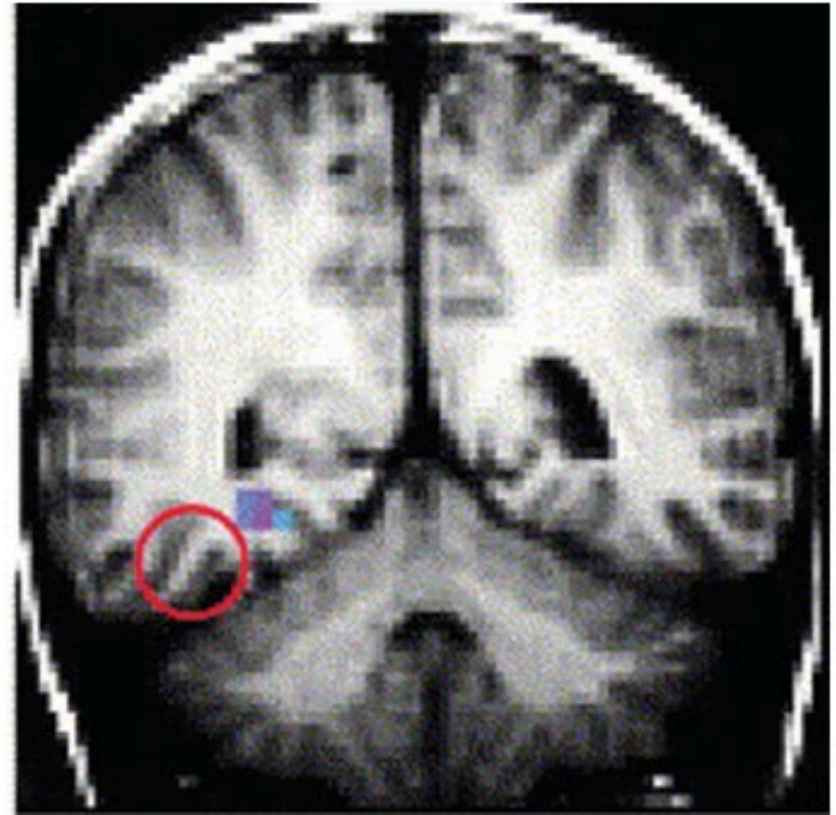


This figure shows the fusiform face area in relation to the striate and extrastriate cortex. The fusiform face area is involved in recognizing individual faces based on experience.

Source: Based on Schultz, R. T. (2005). Developmental deficits in social perception in autism: the role of the amygdala and fusiform face area. *International Journal of Developmental Neuroscience*, 23(2–3), 125–141.



Control



Autistic

FIGURE 16.2 Fusiform Face Area and Autism. The scans show activation of the fusiform face area of control subjects but not of autistic subjects while looking at pictures of human faces.

Autism Spectrum Disorder Possible causes

- Behavioral characteristics appear to be exaggerations of male brain traits
 - Largely controlled by exposure to prenatal androgens
- Changes in neuropeptide signaling may also be involved
 - Lower levels of oxytocin
- New treatment approaches focus on brain-based changes

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Which publications would you like to receive? *

- ☐ Harvard Medicine magazine (monthly)
☐ Harvard Medicine News (weekly)

Cracking the Fever-Autism Mystery

Study suggests fever-induced immune molecule can influence social behavior in mice

By ANNE TRAFTON | December 18, 2019 | [Research](#)



Attention-Deficit/Hyperactivity Disorder

(1 of 2)

How do you define ADHD?

When is it diagnosed?

Attention-Deficit/Hyperactivity Disorder

(1 of 2)

Symptoms

- Appear so often that they interfere with ability to learn
 - Can affect individuals in childhood and adulthood
- Six or more of nine symptoms of inattention and/or hyperactivity and impulsivity for at least six months

Inattentive type – six (or five for people over 17 years) of the following symptoms occur frequently:

- Doesn't pay close attention to details or makes careless mistakes in school or job tasks.
- Has problems staying focused on tasks or activities, such as during lectures, conversations or long reading.
- Does not seem to listen when spoken to (i.e., seems to be elsewhere).
- Does not follow through on instructions and doesn't complete schoolwork, chores or job duties (may start tasks but quickly loses focus).
- Has problems organizing tasks and work (for instance, does not manage time well; has messy, disorganized work; misses deadlines).
- Avoids or dislikes tasks that require sustained mental effort, such as preparing reports and completing forms.
- Often loses things needed for tasks or daily life, such as school papers, books, keys, wallet, cell phone and eyeglasses.
- Is easily distracted.
- Forgets daily tasks, such as doing chores and running errands. Older teens and adults may forget to return phone calls, pay bills and keep appointments.

Hyperactive/impulsive type – six (or five for people over 17 years) of the following symptoms occur frequently:

- Fidgets with or taps hands or feet, or squirms in seat.
- Not able to stay seated (in classroom, workplace).
- Runs about or climbs where it is inappropriate.
- Unable to play or do leisure activities quietly.
- Always "on the go," as if driven by a motor.
- Talks too much.
- Blurts out an answer before a question has been finished (for instance may finish people's sentences, can't wait to speak in conversations).
- Has difficulty waiting his or her turn, such as while waiting in line.
- Interrupts or intrudes on others (for instance, cuts into conversations, games or activities, or starts using other people's things without permission). Older teens and adults may take over what others are doing.

Attention-Deficit/Hyperactivity Disorder

(1 of 2)

Symptoms

- Approximately 5 percent of children; boys are about ten times more likely to be diagnosed
- Often associated with aggression, conduct disorder, learning disabilities, depression, anxiety, and low self-esteem

BROKEN PROMISES

Adults with ADHD were nearly **2X** more likely to have been divorced*1

The consequences may be serious.
Screen for ADHD.

Find out more at
www.consequencesofadhd.com
and download patient support materials,
coupons, and adult screening tools.

*Results from a population survey of 500 ADHD adults and 501 gender- and age-matched non-ADHD adults which investigated characteristics of ADHD and its impact on education, employment, socialization, and personal outlook.

Reference: 1. Biederman J, Faraone SV, Spencer TJ, et al. Functional impairments in adults with self-reports of diagnosed ADHD: a controlled study of 1001 adults in the community. *J Clin Psychiatry*. 2006;67:524-540.

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A1410

11/06

"I'm Depressed..."

Could it be ADHD?
ADHD was found in
32% of adults with a
depressive disorder*1

**Look for ADHD in
patients who present with
depression.**

Visit www.depressionandadhd.com
for patient education kits
and adult screening tools.

*From a retrospective survey assessing the prevalence, comorbidity, and impairment of adult ADHD in 3199 adults, age 18 to 44. Depressive disorder includes major depressive disorder and dysthymia.

Reference: 1. Kessler RC, Adler L, Barkley R, et al. The prevalence and correlates of adult ADHD in the United States: results from the National Comorbidity Survey Replication. *Am J Psychiatry*. 2006;163:716-723.

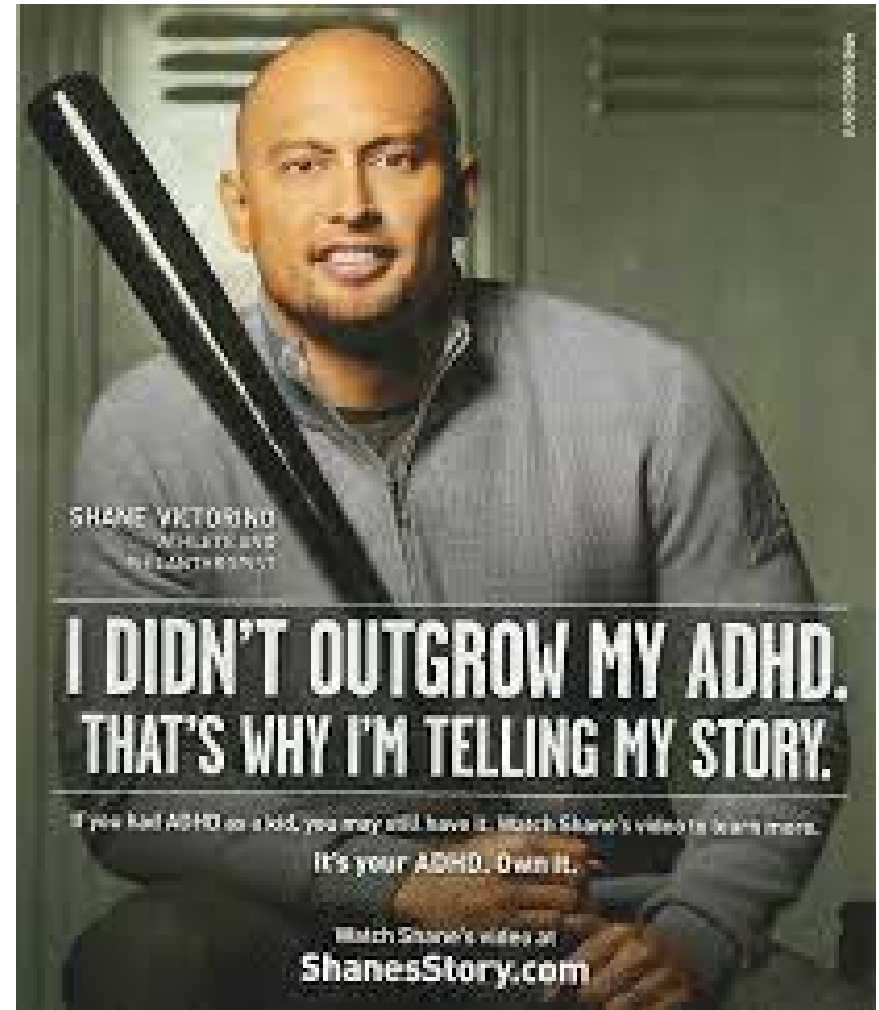
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A1411

11/06



SHANE VICTORINO
ATHLETE AND
PHILANTHROPIST

**I DIDN'T OUTGROW MY ADHD.
THAT'S WHY I'M TELLING MY STORY.**

If you had ADHD as a kid, you may still have it. Watch Shane's video to learn more.
It's your ADHD. Own it.

Watch Shane's video on
ShanesStory.com

Sex differences 2.5 during childhood

Adulthood it is 1:1

The rate is twice the amount in the US and Israel

The U.S. has the highest rate of diagnosed ADHD at 11% compared to a rate of around 5% in countries such as Brazil, China, and Europe. The number is even higher among teenage boys, with figures as high as 20%, according to the CDC's latest report.

Is There An ADHD Epidemic in Major League Baseball?

0 COMMENTS

Eight percent of major league players have been diagnosed with ADHD and prescribed stimulants like Adderall and Ritalin. Critics charge the numbers are much higher in baseball than in the general population. In response, MLB tightened rules for obtaining permission to take the stimulants, which are otherwise banned. We investigate and separate fact from fiction.

By Wendy Thurm | Jun 29, 2012, 11:56am EDT |

f t SHARE



Discussion



Since 2007, more than 100 major leaguers have been granted exemptions each season in order to take drugs like Adderall and Ritalin, stimulants that help ADHD sufferers focus for long periods of time. And it's been growing slightly every season, from 103 in 2007 to 111 in 2011. That's about eight percent of all major leaguers.

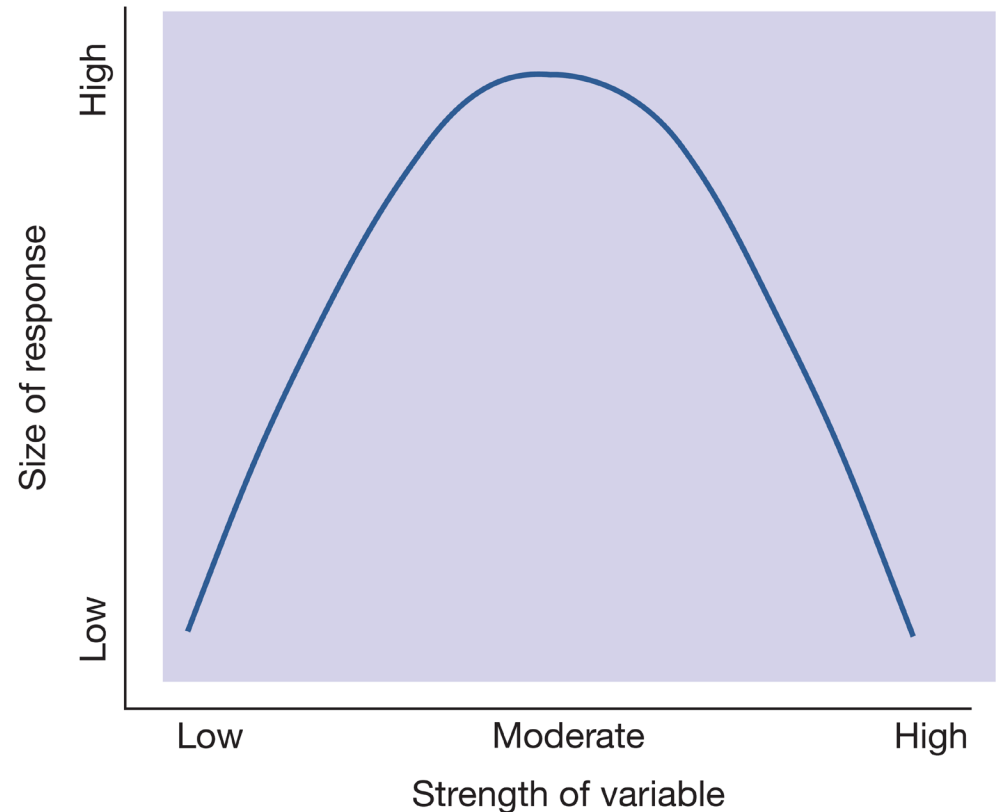
ADHD Possible Causes

- Hereditary factors
- Resemble symptoms produced by damage to the prefrontal cortex
- Hypothesis that it is caused by underactivity of dopaminergic transmission
 - Dose-response curve for the effects of methylphenidate follow inverted U-shaped function

Figure 17.3

An Inverted U Curve and dopamine levels

The graph illustrates an inverted U-curve function, in which low and high values of the variable on the horizontal axis are associated with low values of the variable on the vertical axis and moderate values are associated with high values. Presumably, the relationship between brain dopamine levels and the symptoms of ADHD follow a function like this one.



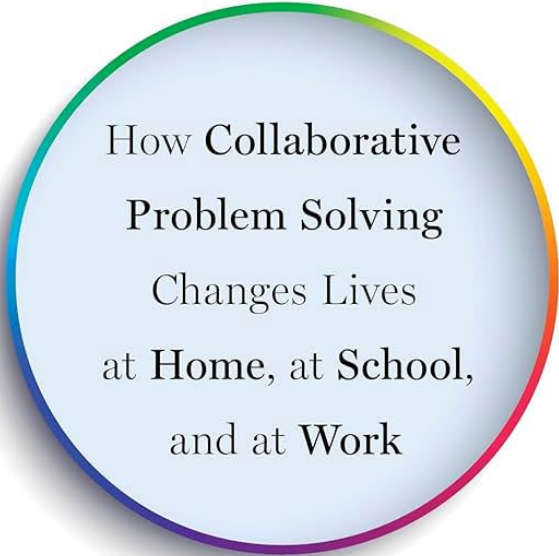
ADHD Treatment

Behavioral therapy and medication can improve the symptoms of ADHD. Studies have found that a combination of behavioral therapy and medication works best for most people, particularly those with moderate to severe ADHD.

Treatment usually consists of teaching parents and teachers how to provide positive feedback for desired behaviors and consequences for negative ones.

Skills not wills

change·*able*



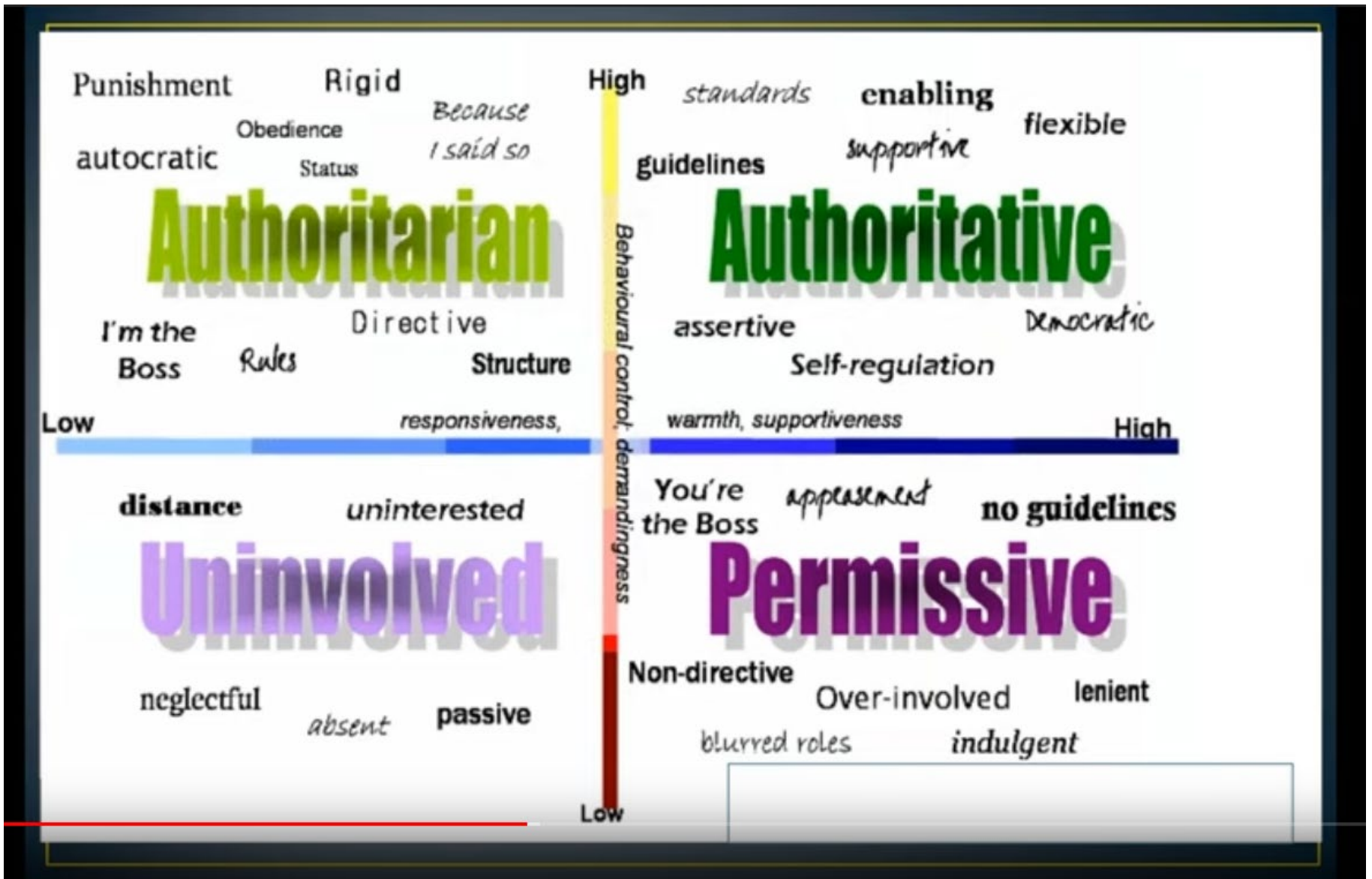
How Collaborative
Problem Solving
Changes Lives
at Home, at School,
and at Work

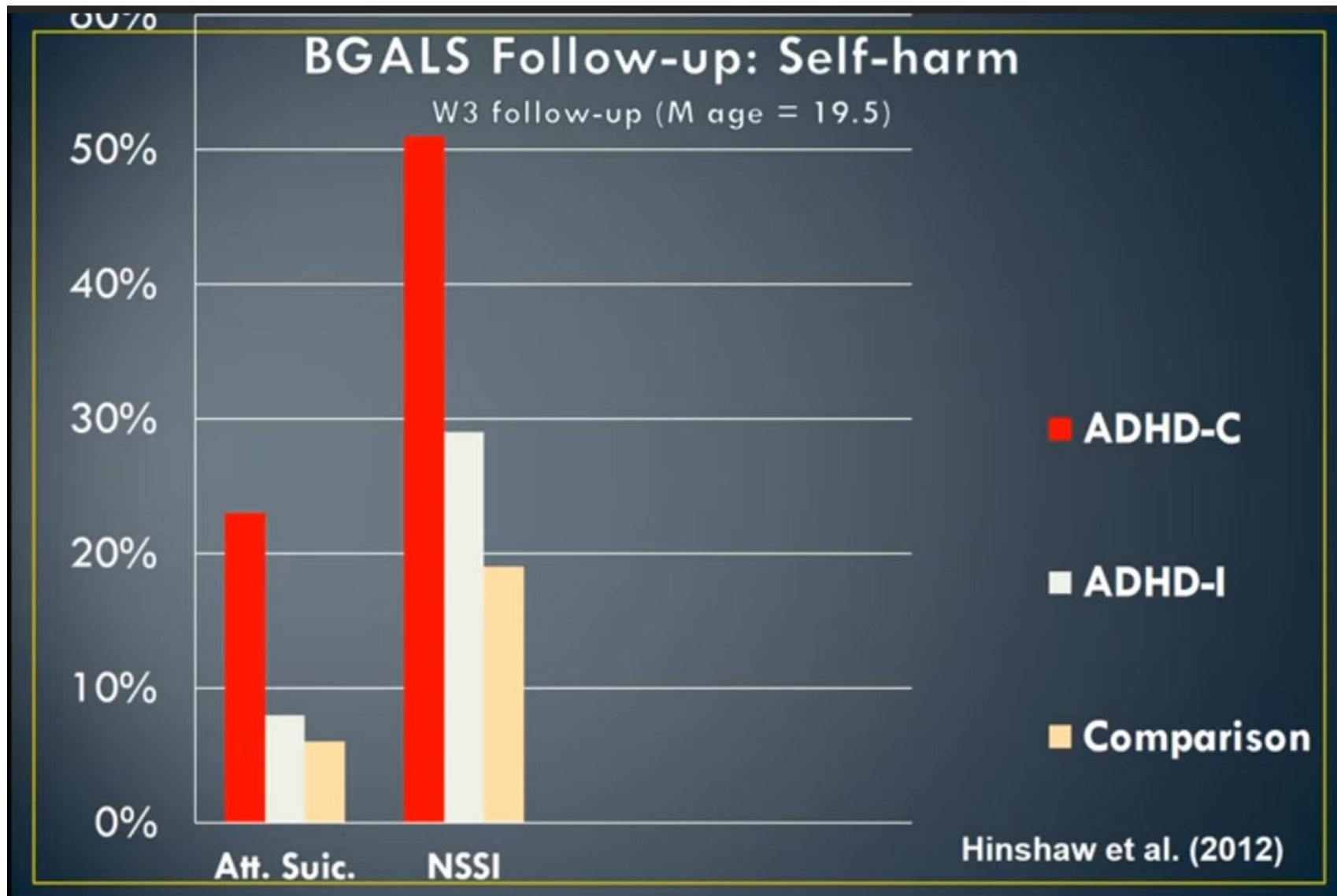
J. Stuart Ablon, PhD

Director of Think:Kids at Massachusetts General Hospital

Parenting

- Make clear schedules.
- Maintain routines.
- Make sure instructions are understood – use simple words and demonstrate.
- Focus on your child when talking to him/her. Avoid multitasking.
- Supervise. Children with ADHD may require more supervision than their peers.
- Maintain communication with the child's teacher.
- Model calm behavior.
- Focus on effort and reward good behavior.

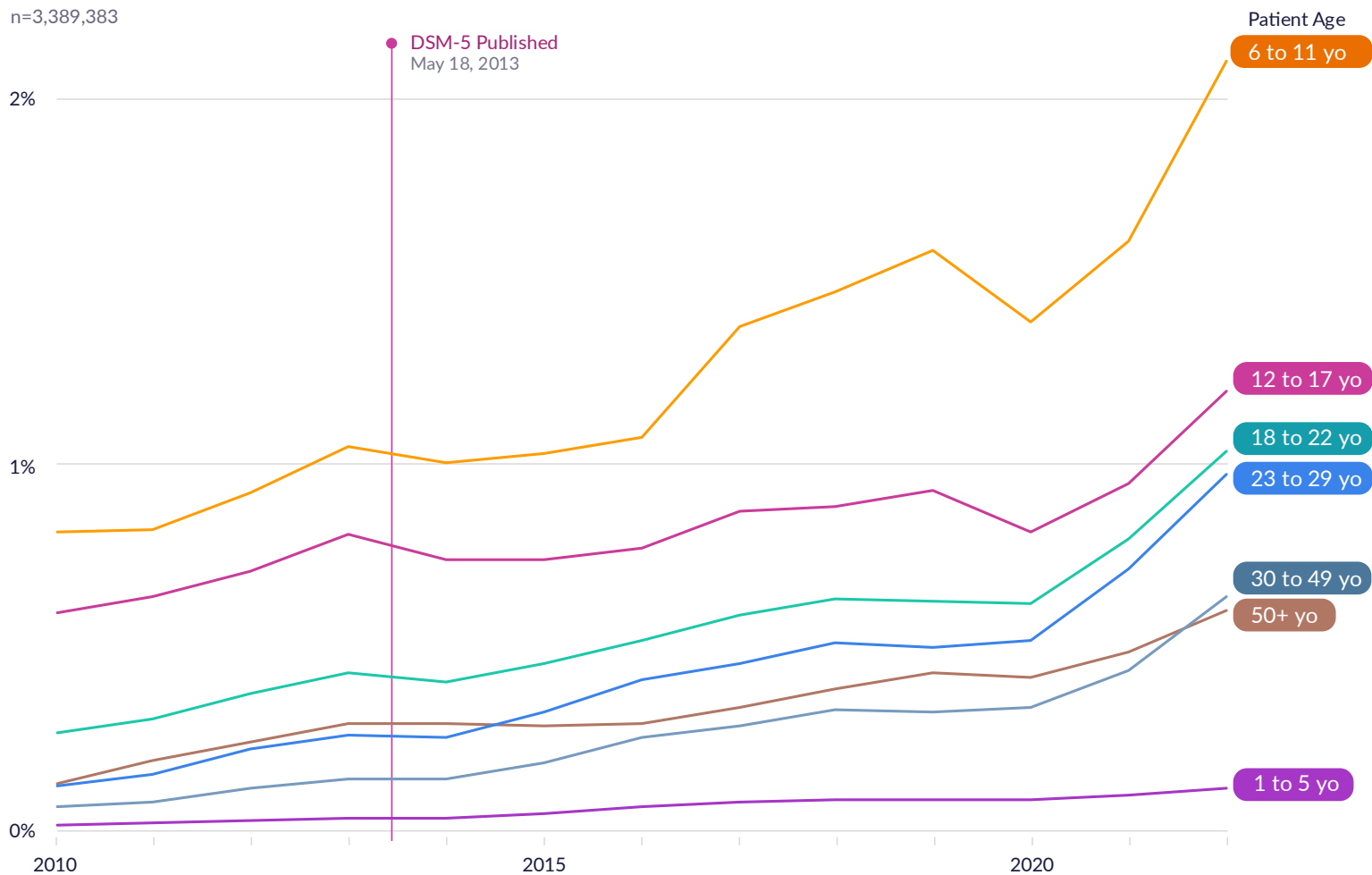




Tidal Wave/ADHD Explosion

National Survey of Children's Health (Visser et al., 2014)
Journal of the American Academy of Child & Adolescent Psychiatry

- **Parent-reported ADHD 'ever diagnosed'**
 - 2003: 7.8%
 - 2007: 9.5%
 - 2012: 11.0%
 - **41% INCREASE IN 9 YEARS**, for all 4-17 year-olds
 - Low-income rates now = middle-class; Black = White
 - Hispanic lower (but fast growing)
- **Medication rates higher, too:**
 - Just under 70% of those 'currently diagnosed' now receive medication
 - Largest medication increases: adolescents, adults



"Age at Initial ADHD Diagnoses Over Time," 2023. EpicResearch.org

These data come from Cosmos, a HIPAA-defined Limited Data Set of more than 183 million patients from 193 Epic organizations including 1,119 hospitals and more than 24,700 clinics, serving patients in all 50 states and Lebanon. This study was completed by two teams that worked independently, each composed of a clinician and research scientists.



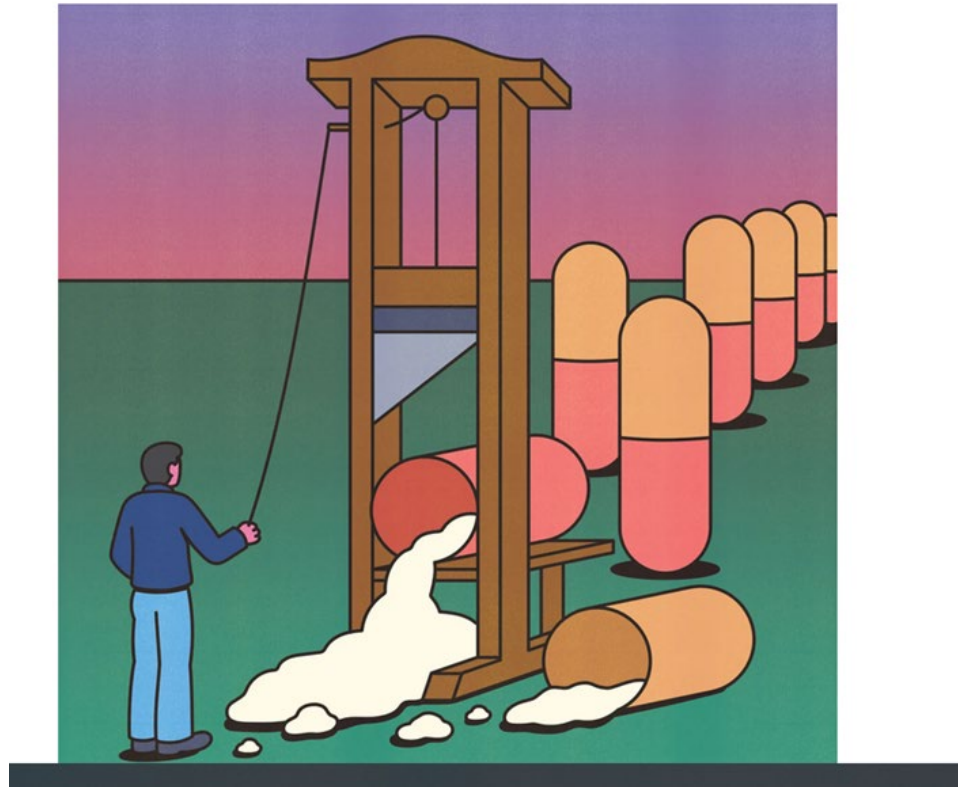
The two teams came to similar conclusions.

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The New York Times

People Can't Get Their A.D.H.D. Medicine, and That's a Sign of a Larger Problem

March 25, 2023



Stress

- **Physiology of the Stress Response**
- **Health Effects of Long-Term Stress**
- **Effects of Stress on the Brain**
- **Stress and Infectious Diseases**

Physiology of the Stress Response (1 of 3)

Learning Objective 17.3 Compare the SAM system and HPA axis in coordinating a stress response.

Stress Response

- A physiological reaction caused by the perception of aversive or threatening situations
 - **Fight-or-flight response**
- Long-term adverse health effects result from the autonomic and endocrine responses



What are some stimuli you find stressful?



Physiology of the Stress Response

Stress Response

- Emotions consist of behavioral, autonomic, and endocrine responses
- Autonomic and endocrine responses can have adverse effects on health

Physiology of the Stress Response:

Sympathetic Adrenal-Medullary System

- Hypothalamus and sympathetic nervous system stimulate the adrenal medulla to release epinephrine and norepinephrine
- These catecholamine hormones initiate a rapid activation of the sympathetic nervous system

Physiology of the Stress Response (3 of 3)

Epinephrine affects glucose metabolism, causing nutrients stored in muscles to become available to provide energy for strenuous exercise

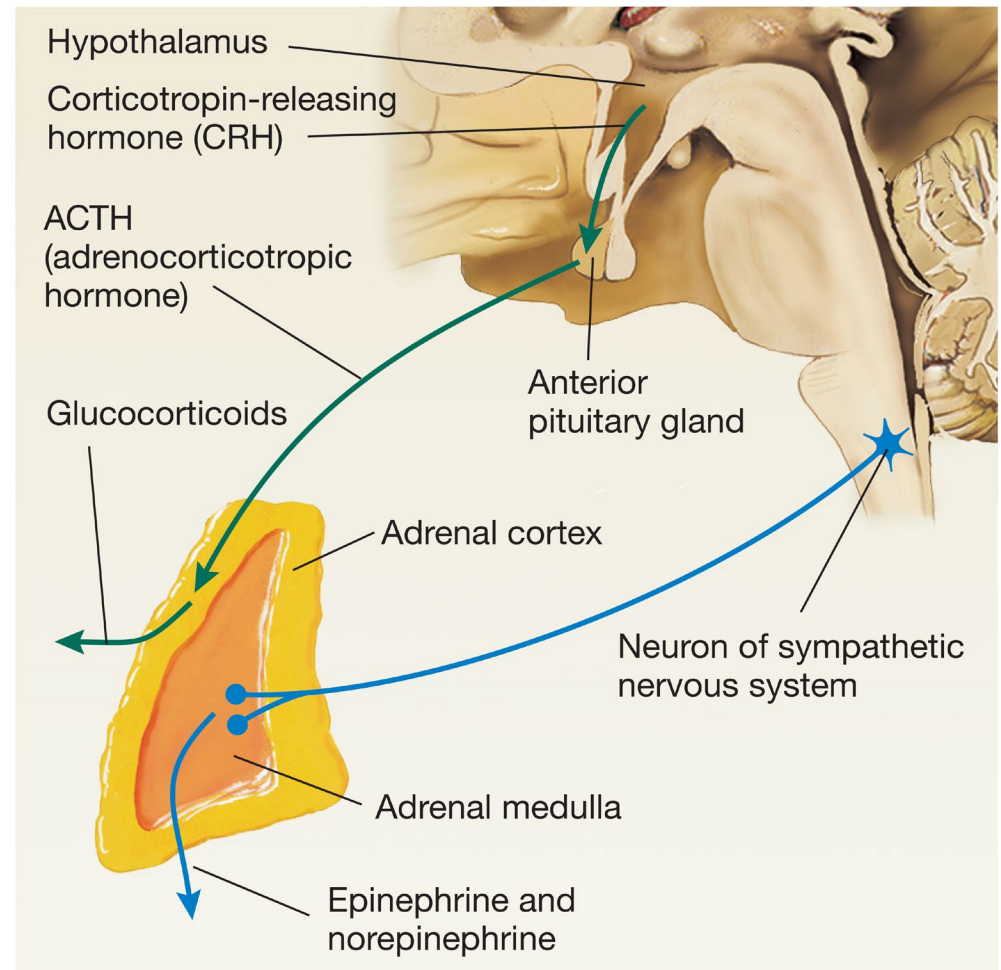
Along with norepinephrine, the hormone also increases blood flow to muscles by increasing output of heart

In doing so, it also increases blood pressure and could lead to cardiovascular disease

Figure 17.4

Control of Secretion of Stress Hormones

The diagram illustrates control of the secretion of glucocorticoids by the adrenal cortex and of catecholamines by the adrenal medulla.



Physiology of the Stress Response (3 of 3)

Hypothalamic Pituitary Adrenal Axis

- **Secretion of Cortisol**, which is a **Glucocorticoid** is a stress-related hormone that has profound effects on glucose metabolism
 - Break down proteins to convert to glucose
 - Helps animals react to stressful situations and survive
 - Make fats available for energy, increase blood flow, stimulate behavioral responsiveness

Health Effects of Long-Term Stress

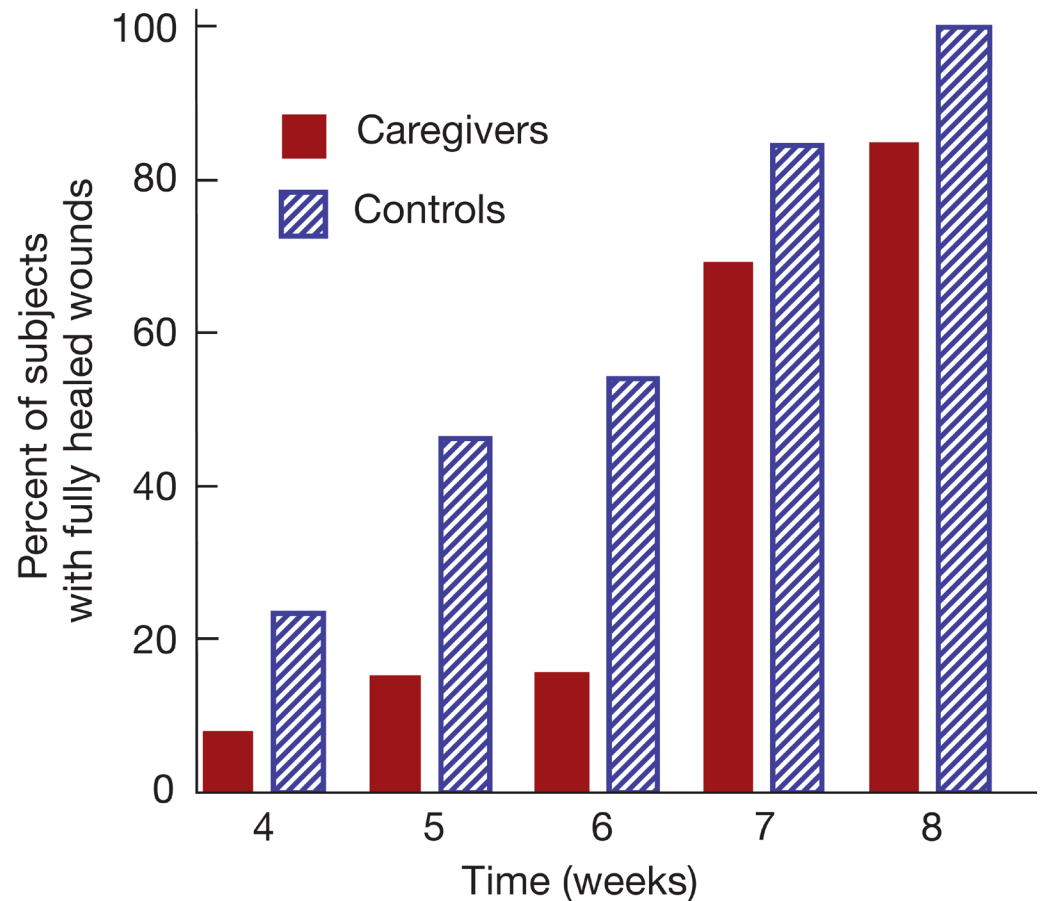
- Often leads to ill health
- Most of the harmful effects of stress are produced by prolonged secretion of glucocorticoids
 - Increased blood pressure, damage to muscle tissue, steroid diabetes, infertility, inhibition of growth, inhibition of inflammatory responses, and suppression of the immune system
- May be due to impaired regulation of the HPA axis

Figure 17.5

Stress and Healing of Wounds

The graph shows the percentage of caregivers and control participants whose wounds had healed as a function of time after the biopsy was performed.

Source: Based on Kiecolt-Glaser, J. K., Marucha, P. T., Malarkey, W. B., Mercado, A. M., & Glaser, R. (1995). Slowing of wound healing by psychological stress. *Lancet*, 346(8984), 1194–1196.



Effects of Stress on the Brain (1 of 3)

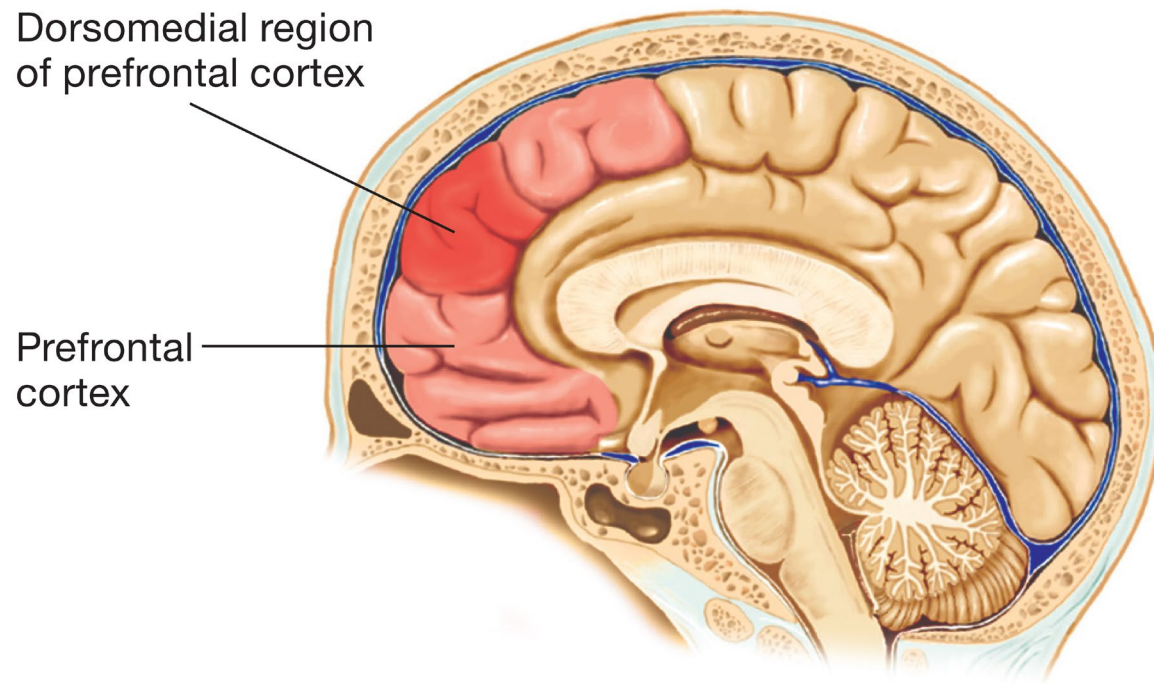
Learning Objective 17.5 Compare the effects of long-term glucocorticoid exposure and early nurturing experiences on the brain in response to stress.

Brain Damage

- Long-term exposure to glucocorticoids destroys neurons located in the hippocampal formation
- Stressors throughout our lives increase the likelihood of memory problems as we grow older
- Intense long-term stress can cause severe brain damage in primates
- The stress of chronic pain has adverse effects on the brain and cognitive behavior

Figure 17.6

Exposure to Early Life Stress Reduces the Volume of the Dorsomedial Prefrontal Cortex



The volume of the dorsomedial prefrontal cortex was reduced 7.2 percent in people who experienced emotional maltreatment during childhood. Source: Based on van Harmelen et al., 2010

Effects of Stress on the Brain (2 of 3)

Prenatal Stress

- Interferes with normal development of hippocampus, leading to long-lasting malfunctions in learning and memory
- Can affect brain development and produce changes for life

Stress and Infectious Diseases

Learning Objective 17.6 Summarize the relationship between stress and infectious disease.

Immune System

- Protects from viruses, microbes, fungi, and parasites
- Stress response can impair functions of the immune system
- Stress-producing events can increase the susceptibility to illness

Figure 17.9

Role of Desirable and Undesirable Events on Susceptibility to Upper Respiratory Infections

The graph shows the mean percentage change in frequency of undesirable and desirable events during the ten-day period preceding the onset of symptoms of upper respiratory

infections. Source: Based on Stone, A. A., Reed, B. R., & Neale, J. M. (1987). Changes in daily event frequency precede episodes of physical symptoms. *Journal of Human Stress*, 13(2), 70–74.

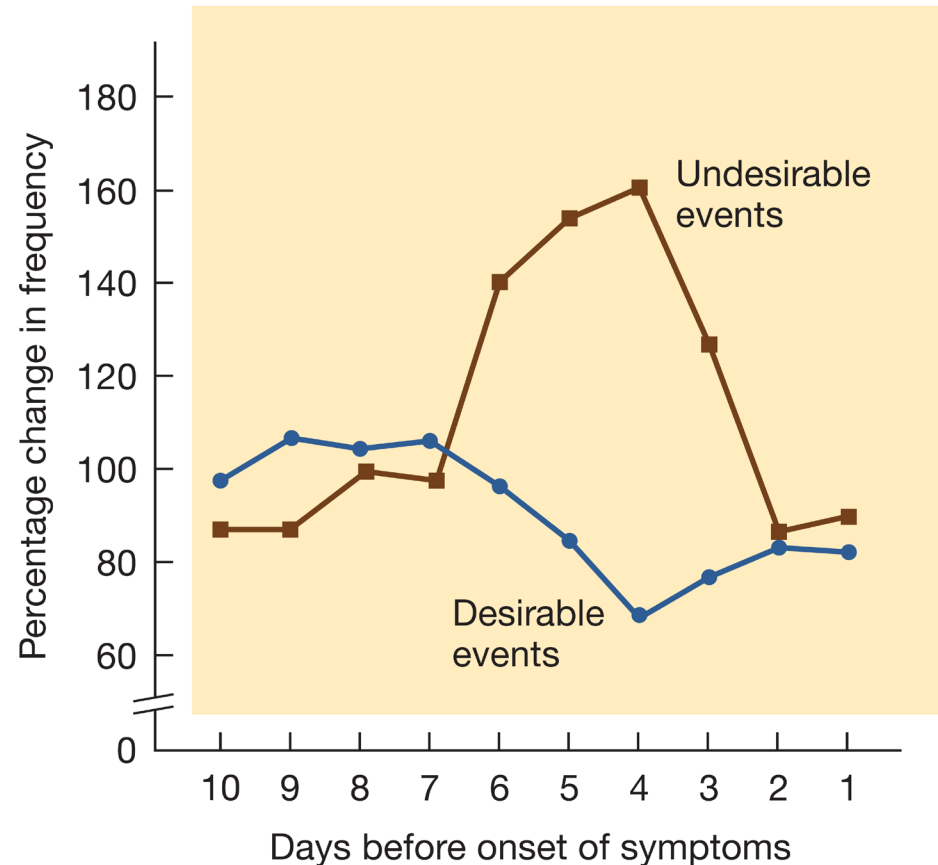


Figure 17.10

Colds and Psychological Stress

The graph shows the percentage of volunteers with colds as a function of an index of psychological stress.

Source: Based on Cohen, S., Tyrrell, D. A., & Smith, A. P. (1991). Psychological stress and susceptibility to the common cold. *New England journal of medicine*, 325(9), 606–612.

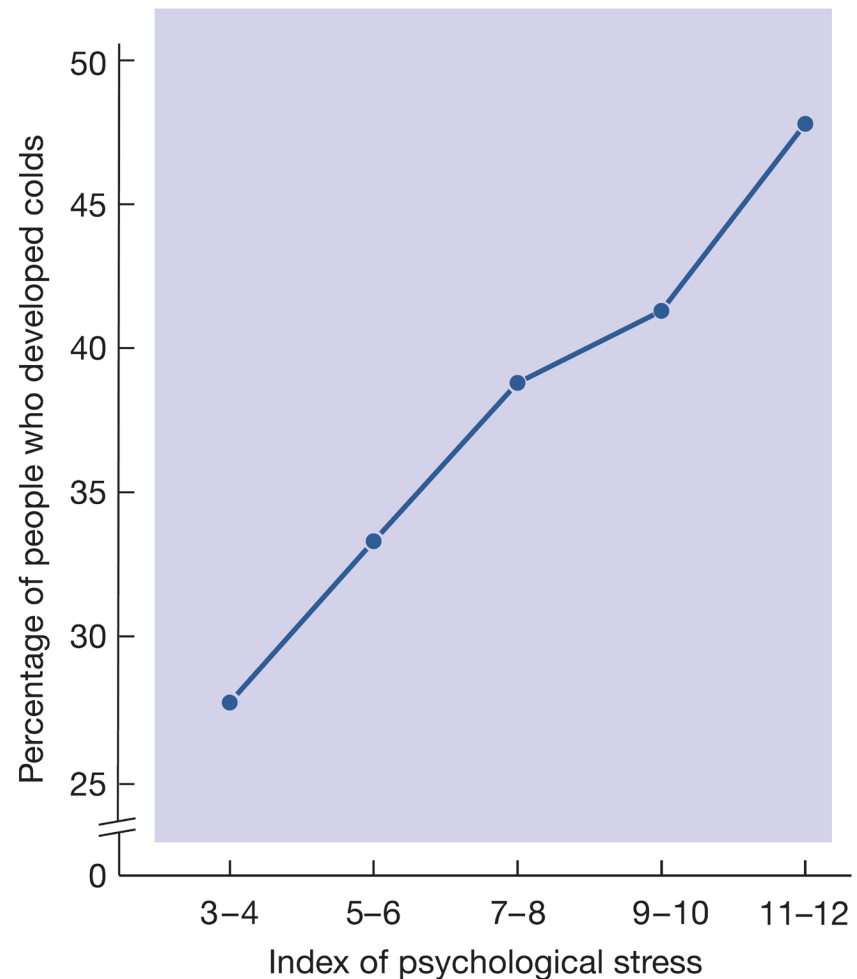
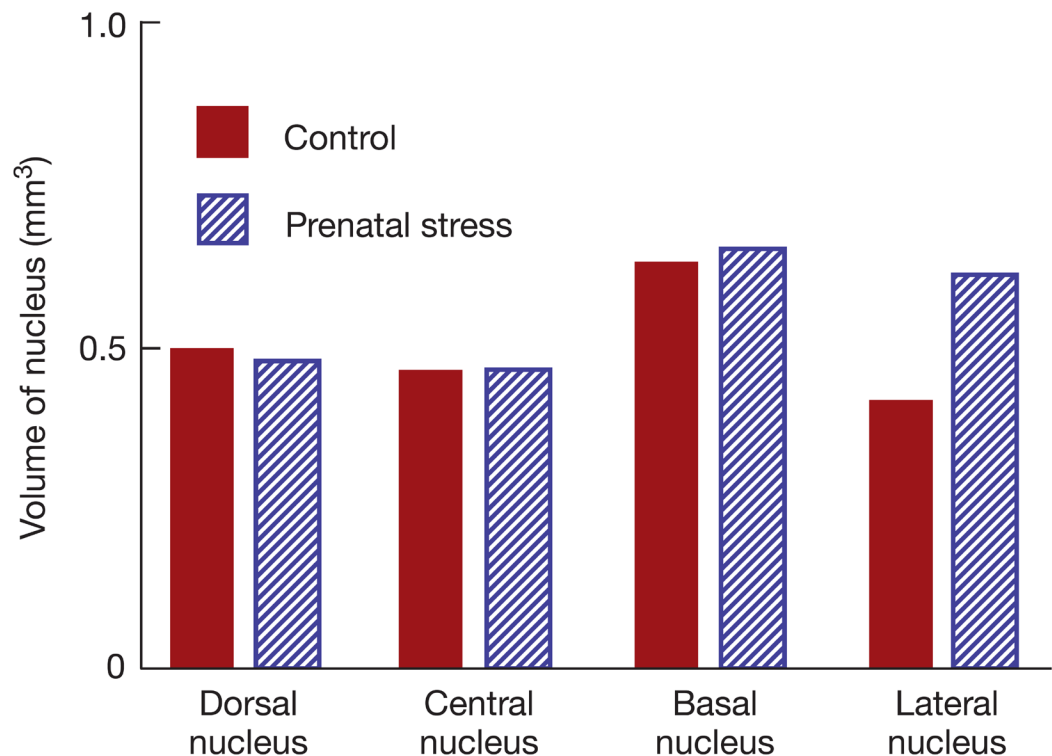


Figure 17.7

Prenatal Stress and the Amygdala

Animals given saline injections

Once a day the last week of gestation





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Developmental Brain Research 148 (2004) 159–167

**DEVELOPMENTAL
BRAIN
RESEARCH**

www.elsevier.com/locate/devbrainres

Research report

Lateral amygdaloid nucleus expansion in adult rats is associated with exposure to prenatal stress[☆]

A.K. Salm^{a,*}, Michelle Pavelko^a, E. Marshall Krouse^a, Wendy Webster^a,
Michał Kraszpulski^{a,b}, Dale L. Birkle^a

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Accepted 29 September 2003

Figure 1

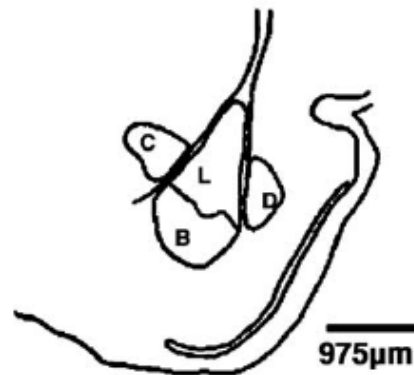
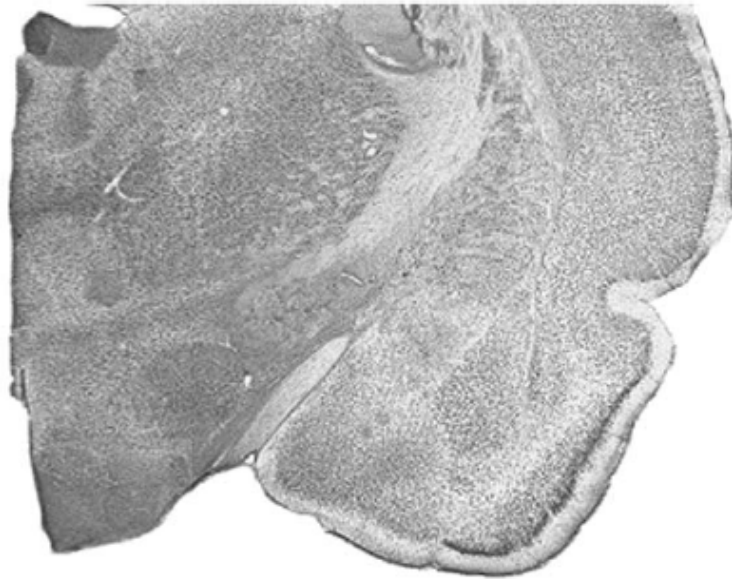


Figure 2

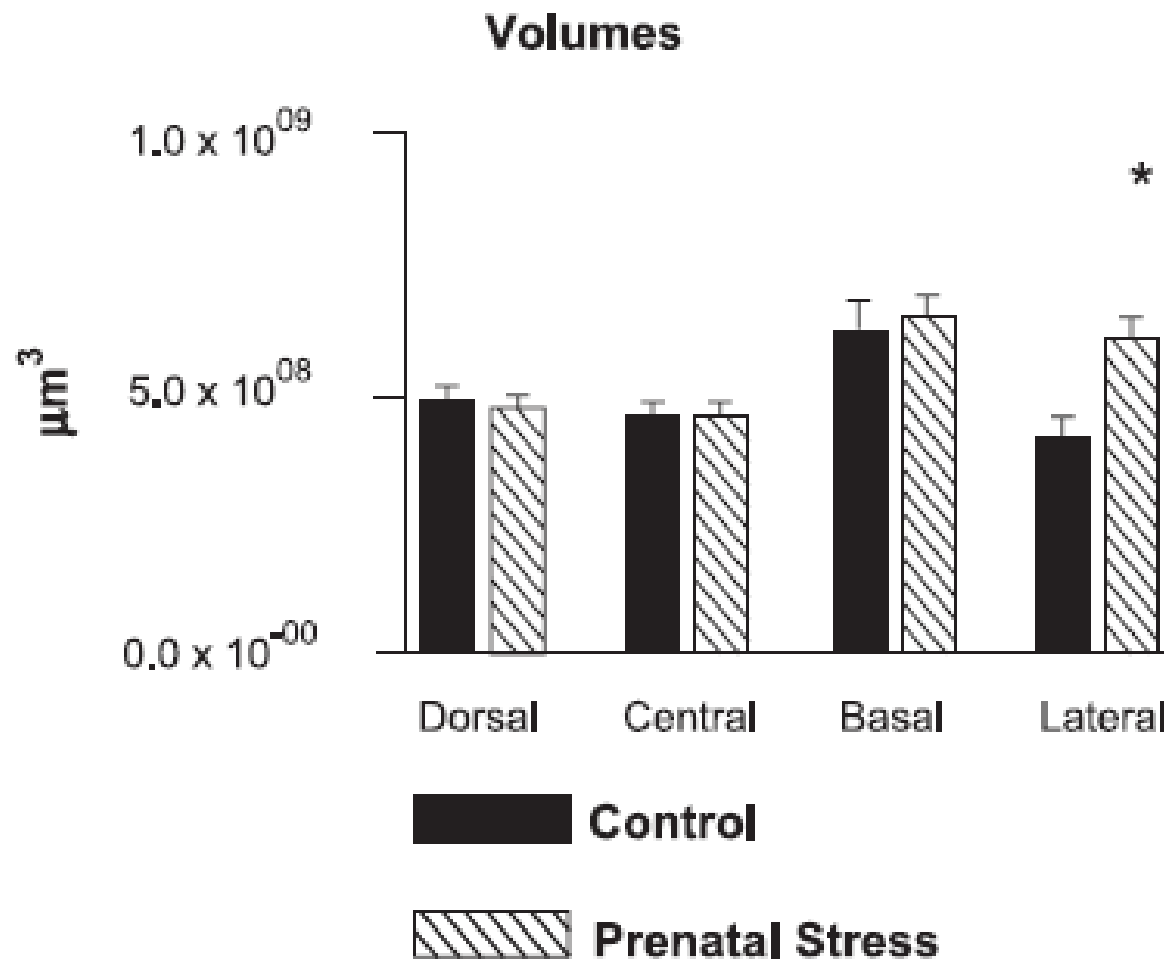


Figure 3

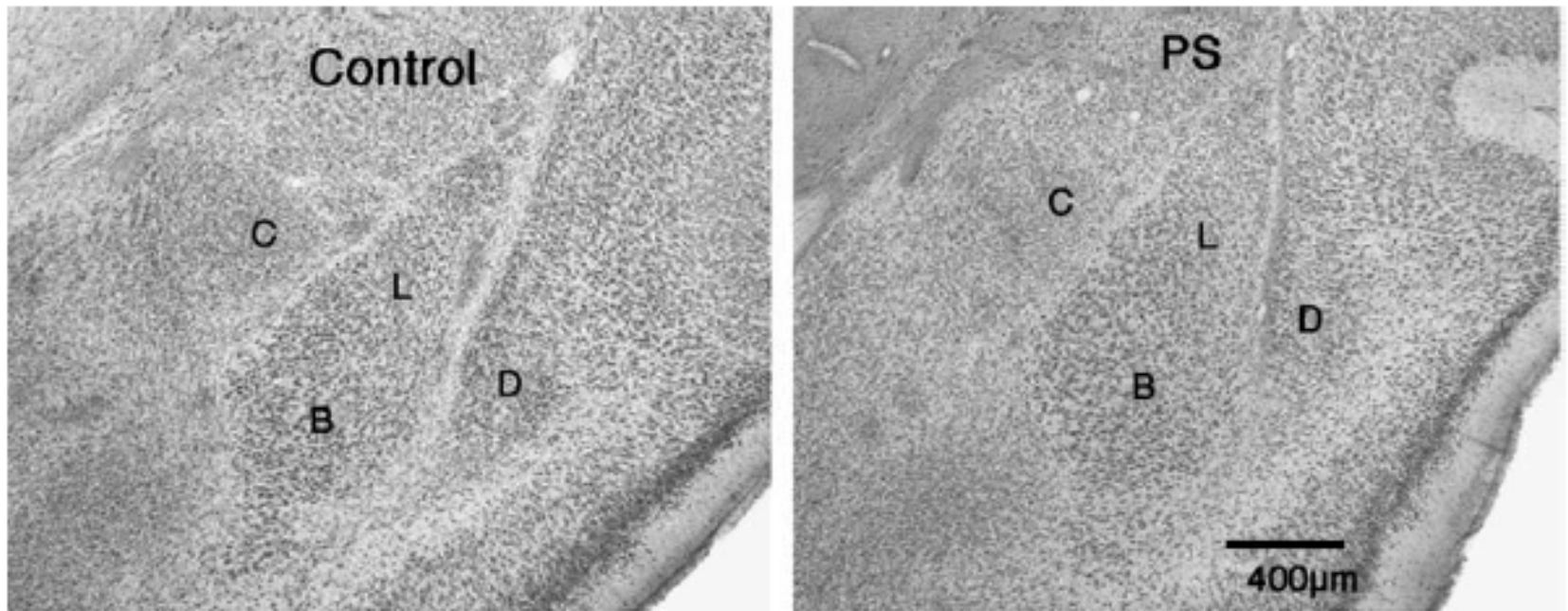


Figure 4

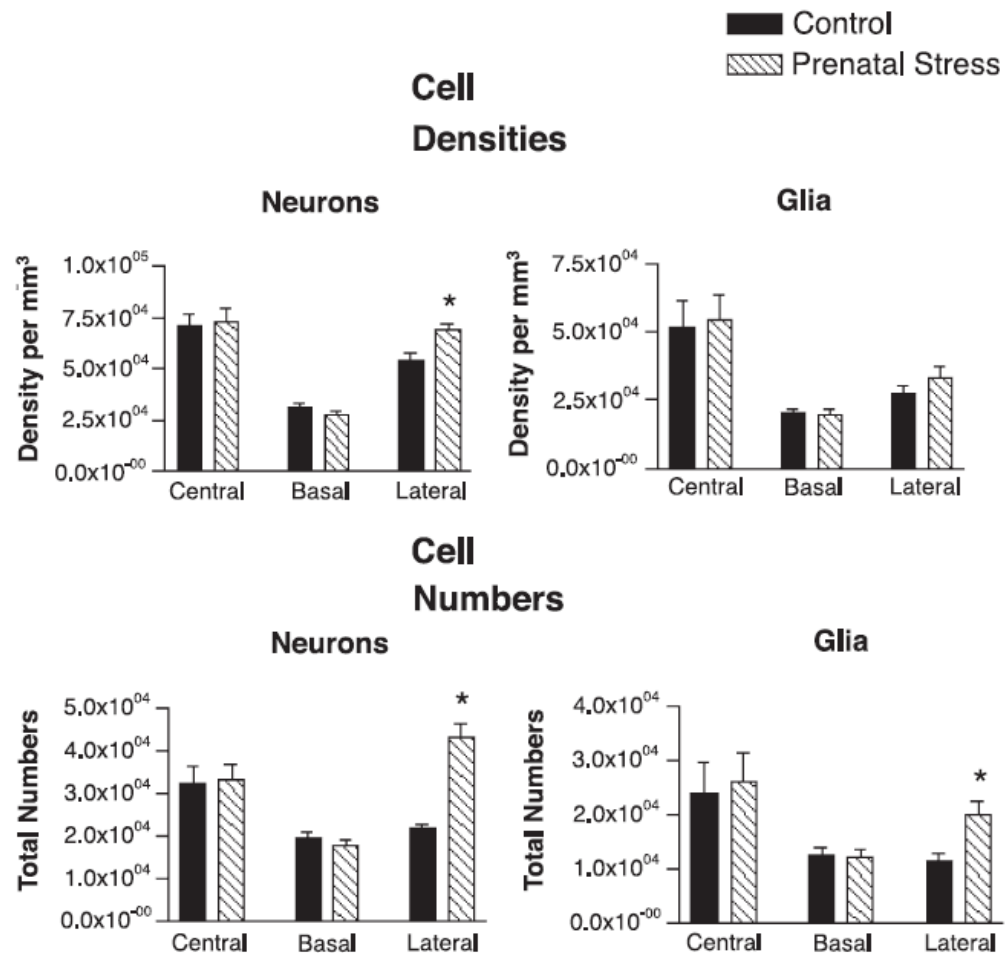
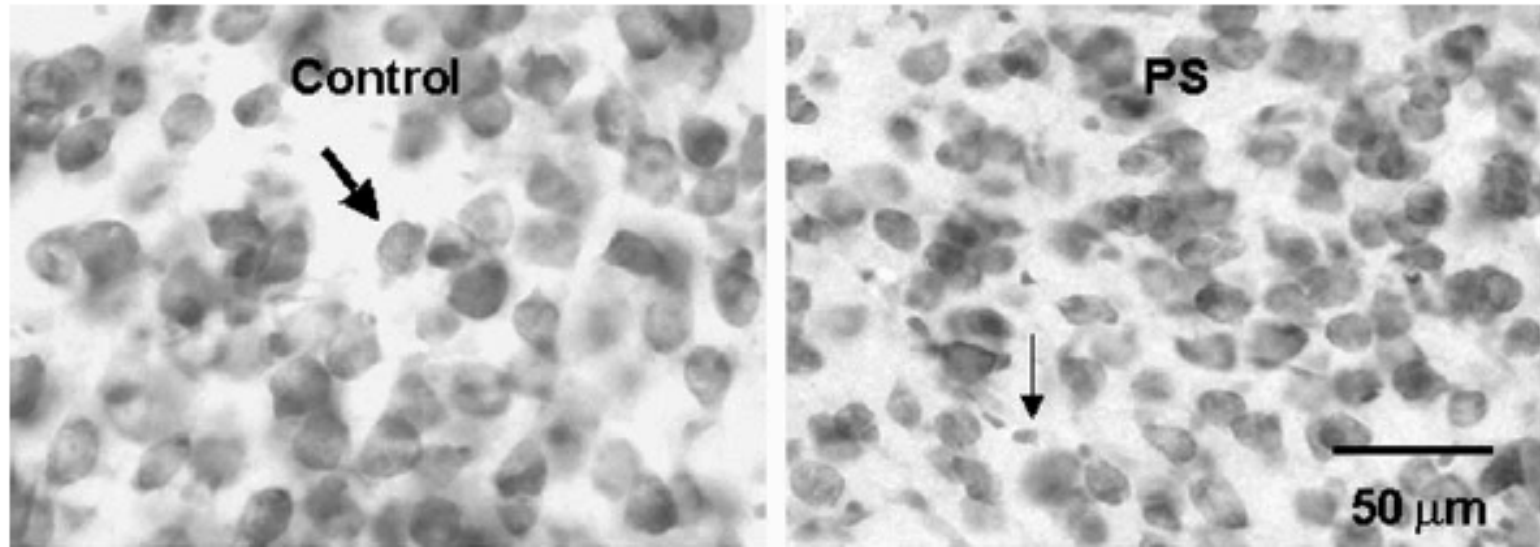


Figure 5



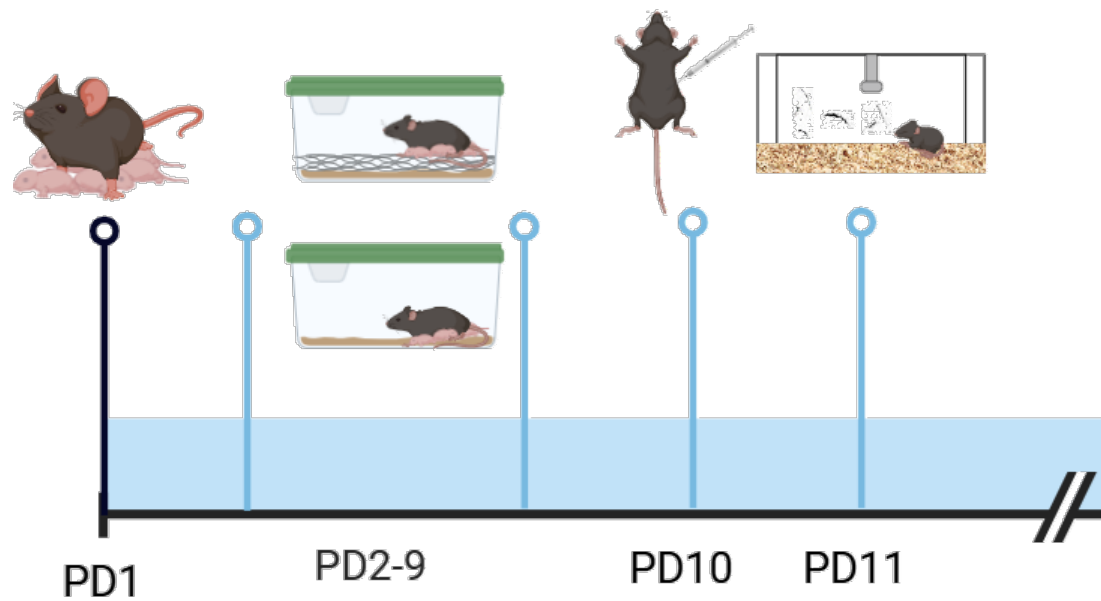
Methods and Materials

Early-Life Stress Simulation

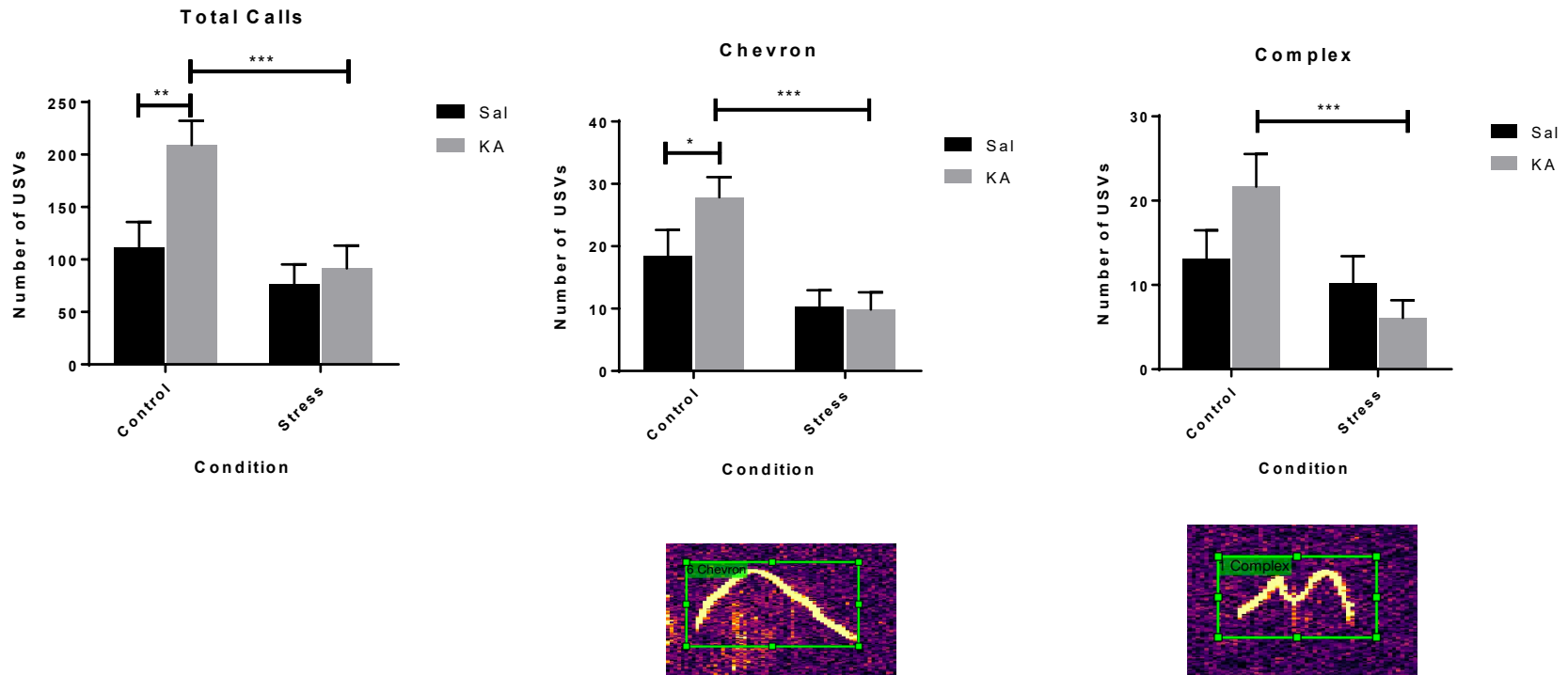
- Limited bedding and nesting (LBN) paradigm
- HPA regulation
- Long Term Consequences



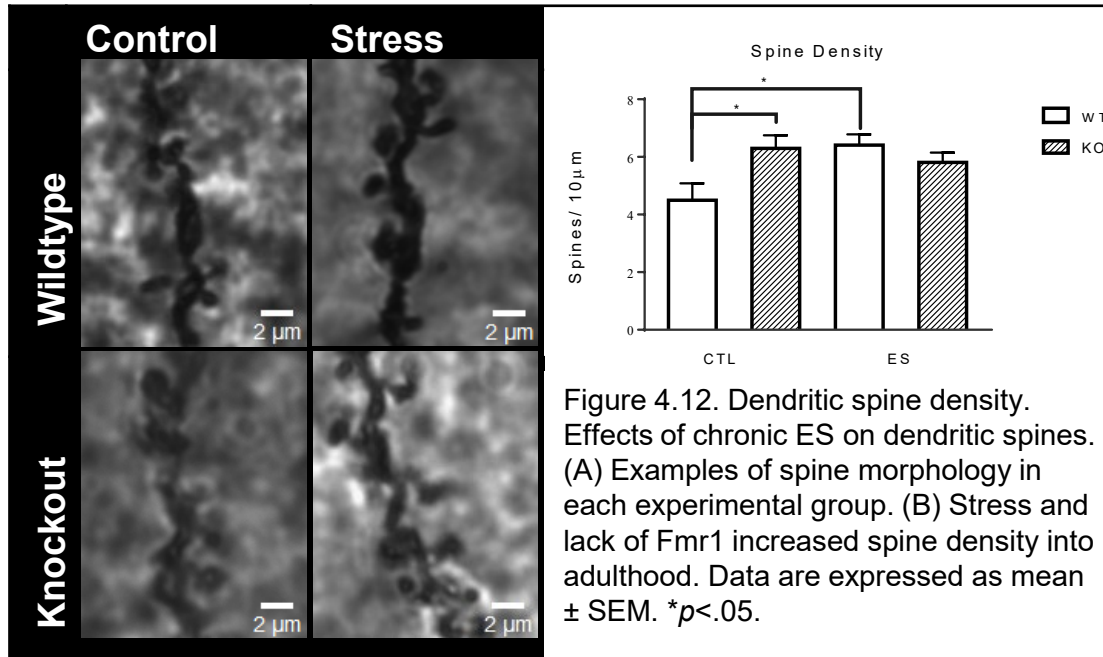
Timeline



Early-life stress results in a decrease in ultrasonic vocalizations



Early-life stress results in an increase in spine density in the medial PFC



Effects of Stress on the Brain:

Good news

Resilience

- Factors such as the presence of various protective hormones and controlled exposure to stress-related stimuli can promote resilience in the event of stress
- Some experiences during early life can reduce reactivity to stressful situations in adulthood

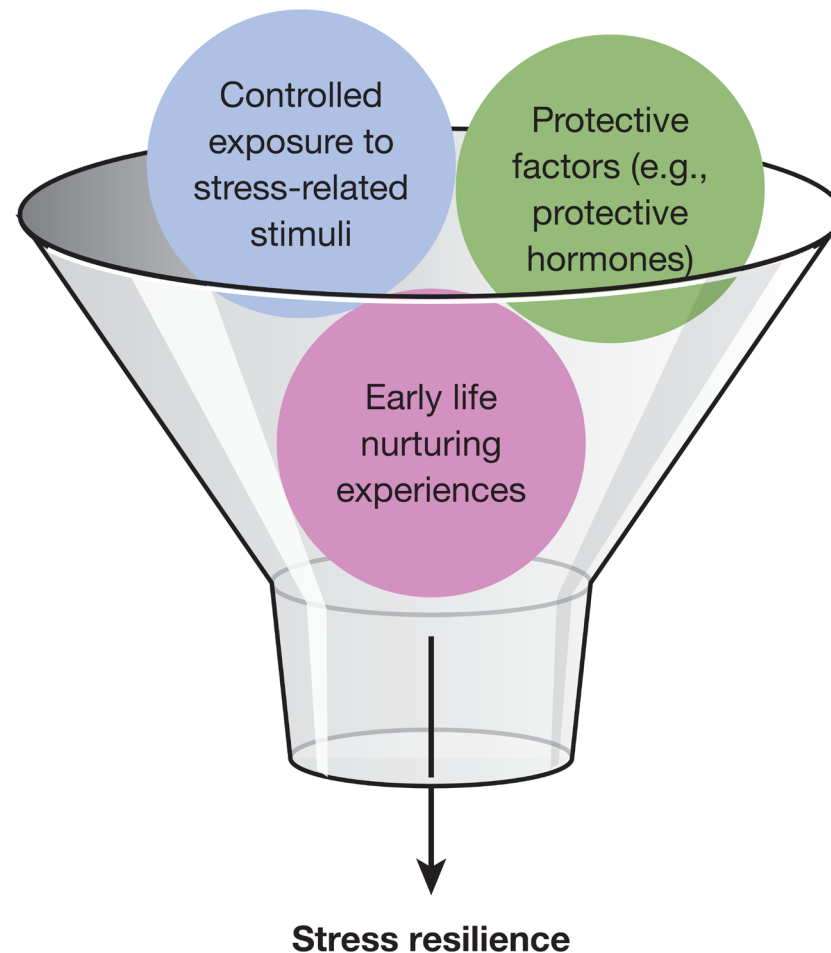
Effects of Stress on the Brain (3 of 3)

Resilience

- Fenoglio et al., (2006) removed rat pups from their cage, handled them for 15 minutes, then returned them to the cage.
- The moms immediately began licking and grooming the pups
- This nurturing behavior activated the CEA, paraventricular nucleus of the hypothalamus, which produced CRH.
 - Reduction of CRH

Figure 17.8

Factors Related to Stress Resilience









Posttraumatic Stress Disorder

- **Symptoms**
- **Possible Causes**
- **Treatment**

Symptoms of PTSD

Likelihood is increased if a traumatic event involves danger or violence from other people

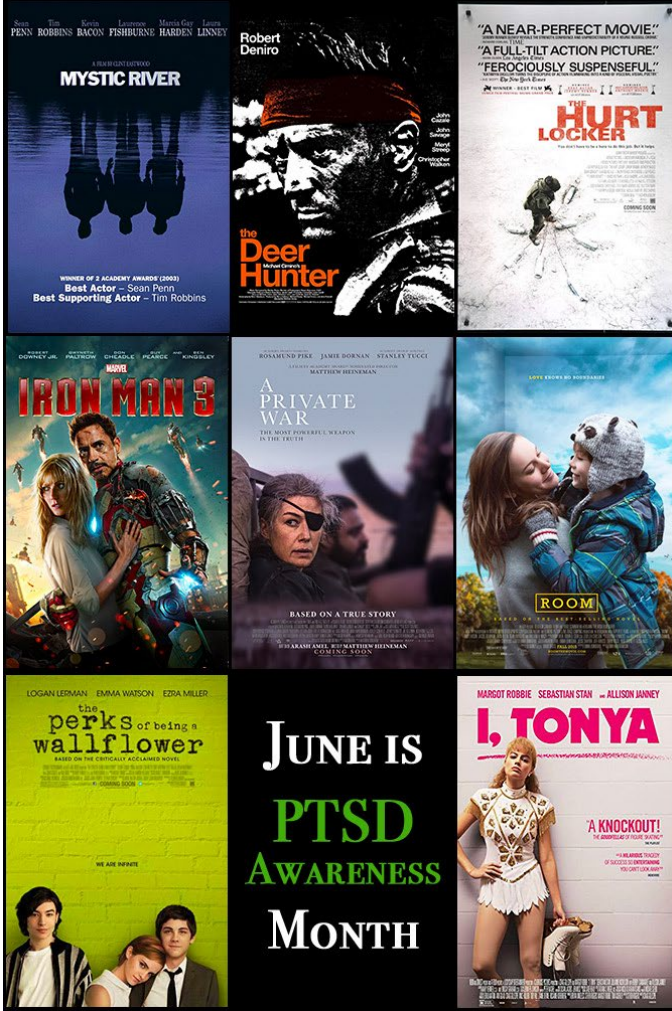
Includes recurrent dreams or recollections of the traumatic event, feelings that the traumatic event is recurring, and intense psychological distress

Impaired mental health functioning

Poor physical health

8 Movies About PTSD

(POST-TRAUMATIC STRESS DISORDER)



DONALD SUTHERLAND MARY TYLER MOORE TIMOTHY HUTTON

ORDINARY PEOPLE

DIRECTED BY
ROBERT REDFORD

“... an intelligent, perceptive, and deeply moving film...”
Roger Ebert, *Chicago Sun-Times*



© A.M.P.A.S.®

WINNER OF FOUR ACADEMY AWARDS®
INCLUDING BEST PICTURE

Genetic and Environmental Factors

- Correlation with the number of traumatic events the person experiences
- The low-activity Met allele in COMT Val¹⁵⁸Met polymorphism is associated with higher availability of dopamine in prefrontal cortex and with higher prevalence of PTSD.
- Shared genetic factors between PTSD and other mental health disorders and behavior tendencies

Figure 17.11

Prevalence of PTSD and Traumatic Events

This graph shows the prevalence of PTSD in survivors of the Rwandan genocide as a function of the number of traumatic events they had suffered.

Source: Kolassa, I. T., Kolassa, S., Ertl, V., Papassotiropoulos, A., & Dominique, J. F. (2010). The risk of posttraumatic stress disorder after trauma depends on traumatic load and the catechol-O-methyltransferase Val158Met polymorphism. *Biological psychiatry*, 67(4), 304–308.

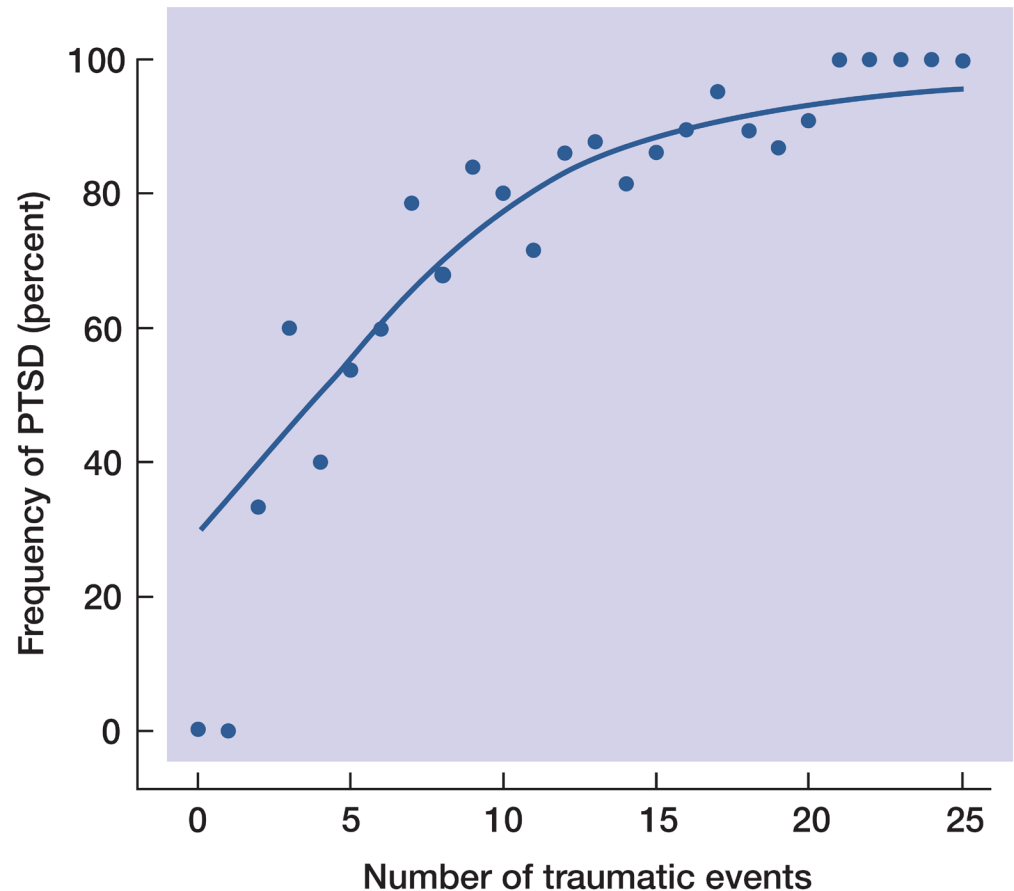
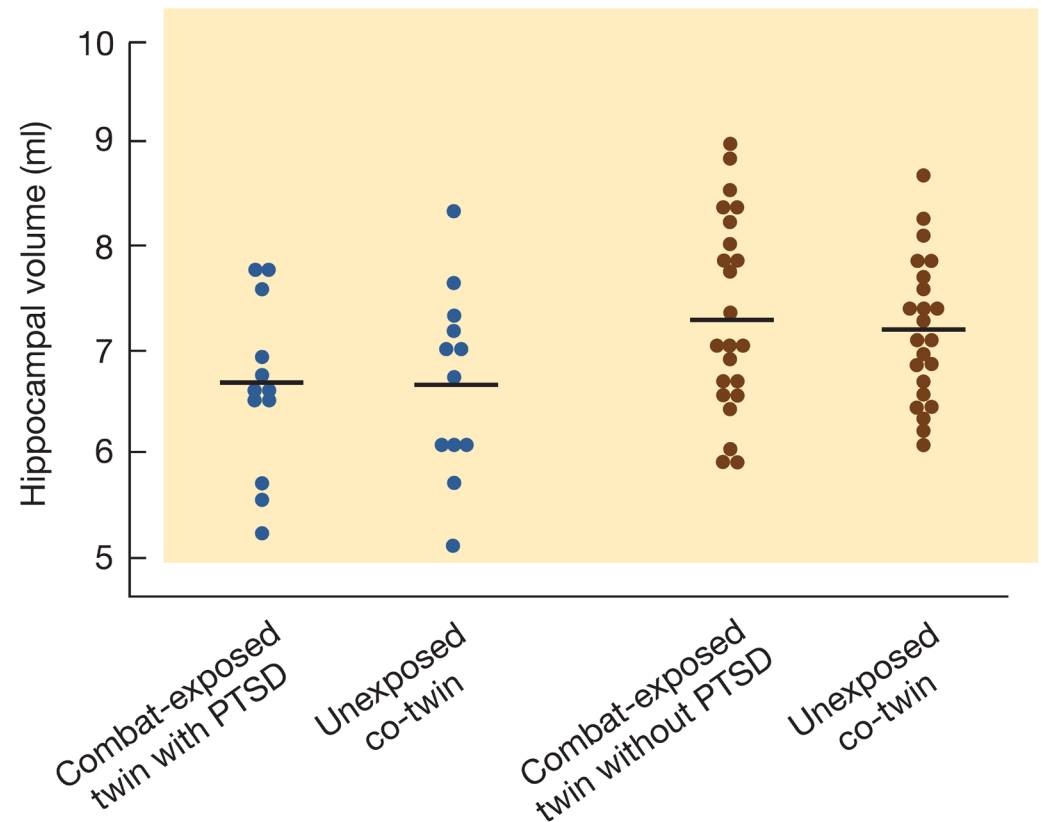


Figure 17.12

Hippocampal Volumes of Pairs of Monozygotic Twins

The graph shows that the size of the hippocampus of twins not exposed to combat was similar to the size of their combat-exposed co-twins whether or not the co-twins had PTSD. These results suggest that hippocampal size is a genetically determined trait that predates the exposure to combat. Source: Gilbertson, M. W., Shenton, M. E., Ciszewski, A., Kasai, K., Lasko, N. B., Orr, S. P., & Pitman, R. K. (2002). Smaller hippocampal volume predicts pathologic vulnerability to psychological trauma. *Nature neuroscience*, 5(11), 1242.



Why would a smaller hippocampus affect the likelihood of developing PTSD? What are some of the treatments for PTSD?

Possible Causes: Hippocampus

- PTSD associated with a smaller hippocampus
 - May predate the exposure to stress
- Aids in distinguishing safe from dangerous contexts
- Prefrontal cortex can exert an inhibitory effect on the amygdala and suppress emotional reactions
- PTSD and TBI co-occur in many individuals

Treatment Options

- Cognitive behavior therapy
- Group therapy
- Antidepressants
- Some treatments focus on preventing PTSD following a traumatic event
 - Administration of cortisol to provide additional negative feedback to the HPA axis to reduce its activity
 - Propranolol-beta adrenergic receptor antagonist
- Stellate Ganglion Block

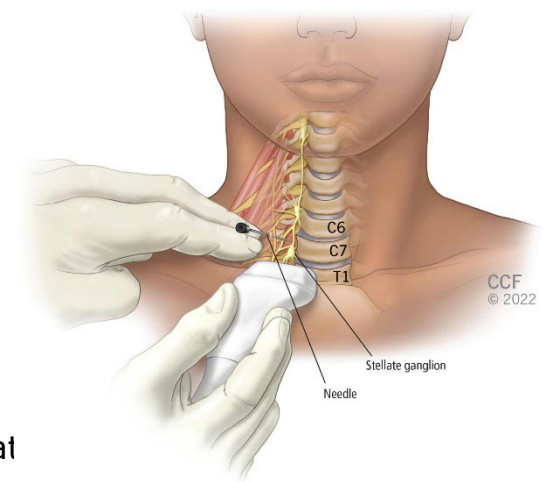
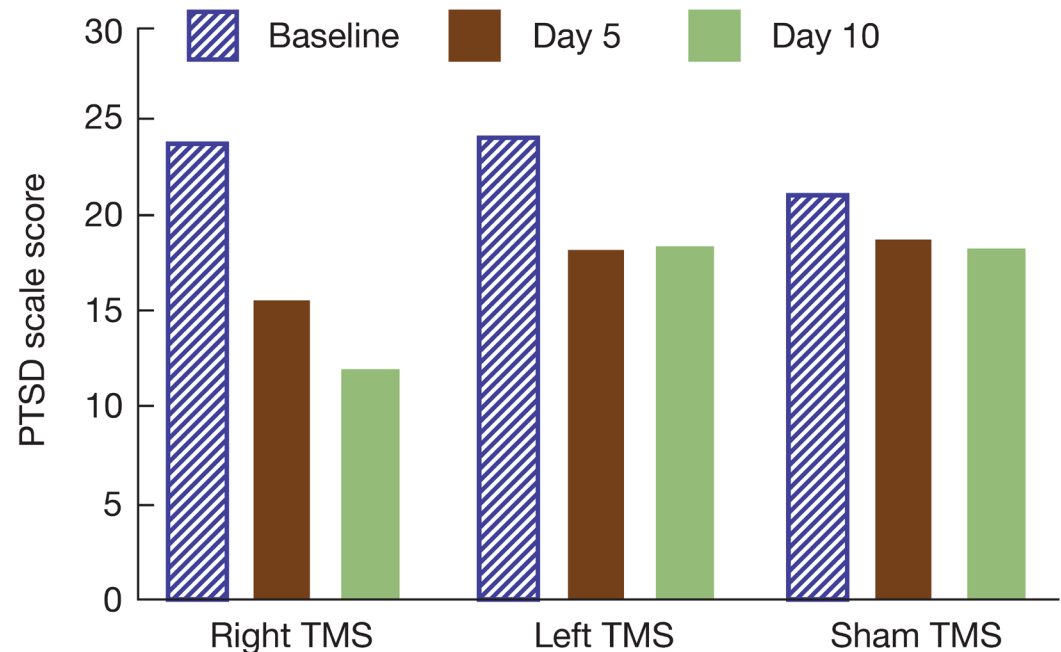


Figure 17.13

Transcranial Magnetic Stimulation and PTSD

The graph shows the effects of TMS of the dorsolateral prefrontal cortex on symptoms of PTSD. Source: Boggio, P. S., Rocha, M., Oliveira, M. O., Fecteau, S., Cohen, R. B., Campanhã, C., . . . & Pascual-Leone, A. (2010). Noninvasive brain stimulation with high-frequency and low-intensity repetitive transcranial magnetic stimulation treatment for posttraumatic stress disorder. *The Journal of clinical psychiatry*, 71(8), 992.



Substance Use Disorders

- **The Role of Reinforcement**
- **Commonly Abused Drugs**
- **Treatment for Substance Use Disorders**

Why do people take drugs?

What is the role of positive reinforcement and negative reinforcement?

The Role of Reinforcement (1 of 5)

Positive Reinforcement

- Drugs that lead to dependency act as a reinforcing stimulus
- Effectiveness is greatest if a reinforcing stimulus occurs immediately after a response
 - Individuals who use drugs recreationally prefer heroin because it has a more rapid effect

Neural Mechanisms

- Drugs trigger the release of dopamine in the nucleus accumbens
- Produce long-term changes in the brain
 - Drugs increase the strength of excitatory synapses on dopaminergic neurons in the VTA
 - Leads to increased activation in regions that receive dopaminergic input from the VTA

Neural Mechanisms

- Synaptic changes that are responsible for compulsive behaviors occur after continued use
 - Occur in the dorsal striatum

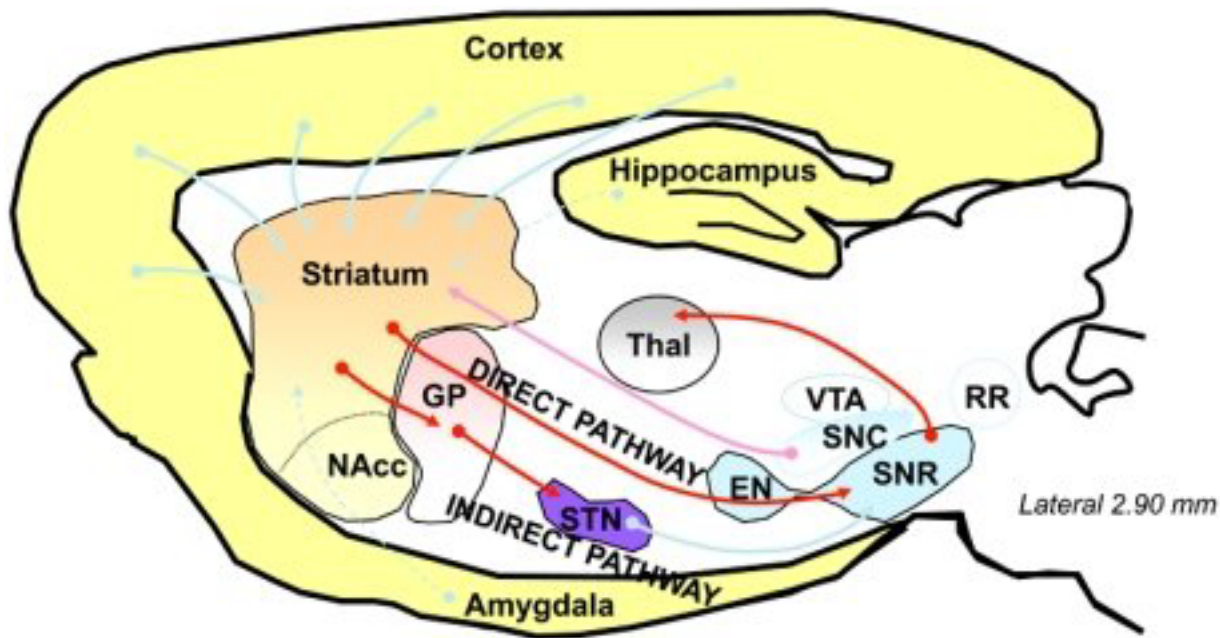


Figure 13.4

A Simple Neural Model of Operant Conditioning

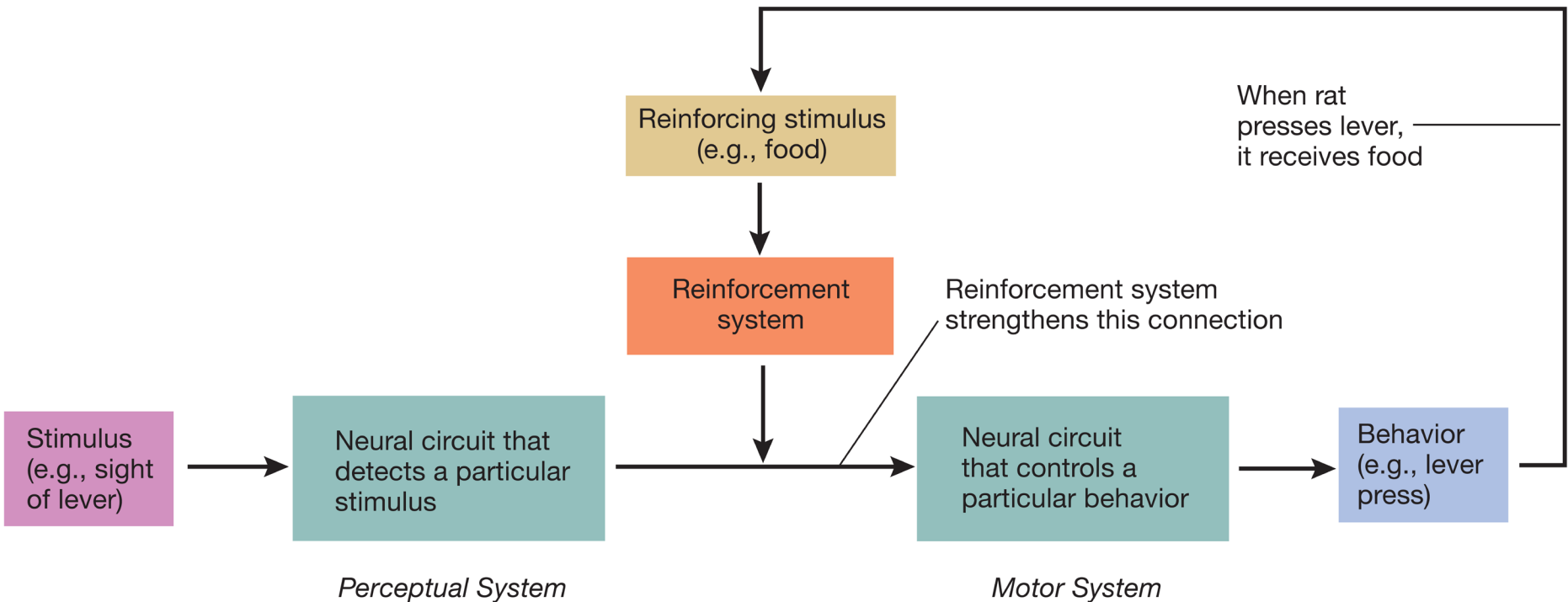
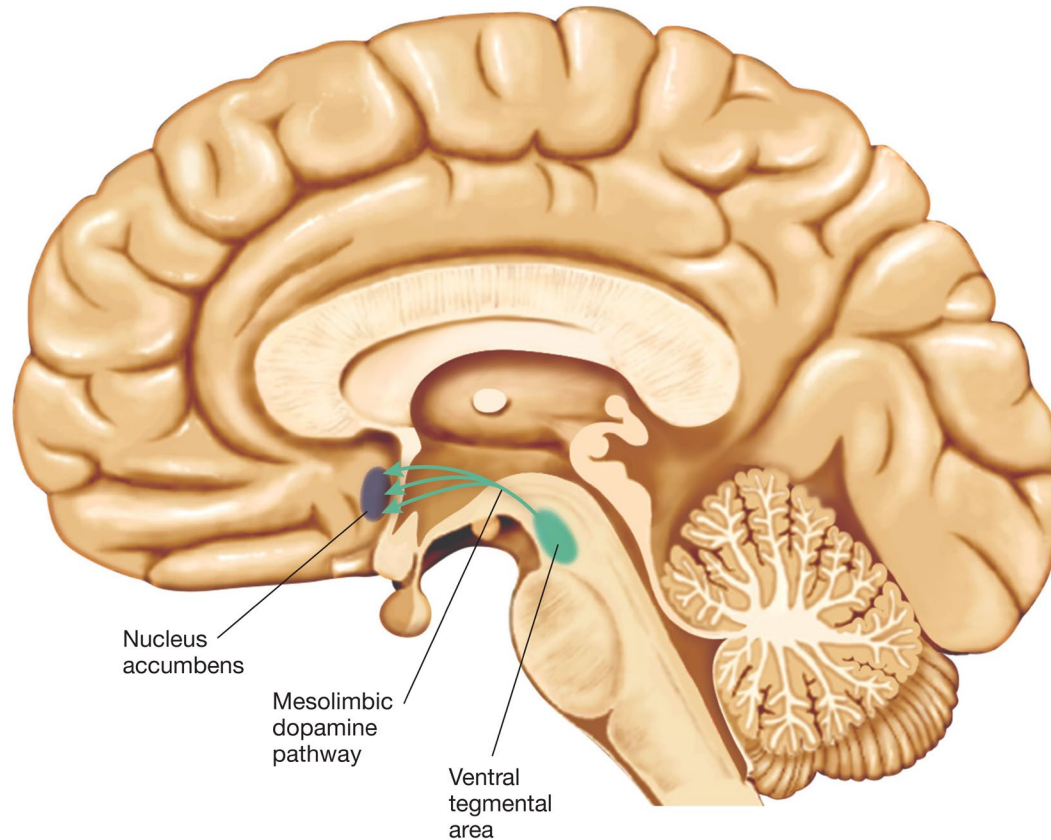


Figure 17.14

Activation of the Mesolimbic Dopamine Pathway Is Necessary for Reinforcement



Negative Reinforcement

- A behavior that turns off (or reduces) an aversive stimulus is reinforced
- People become physically dependent by showing tolerance and withdrawal symptoms
 - Compensate for the unusual condition of the repeated presence of a drug in the body
- Maintain pattern of repeated drug use



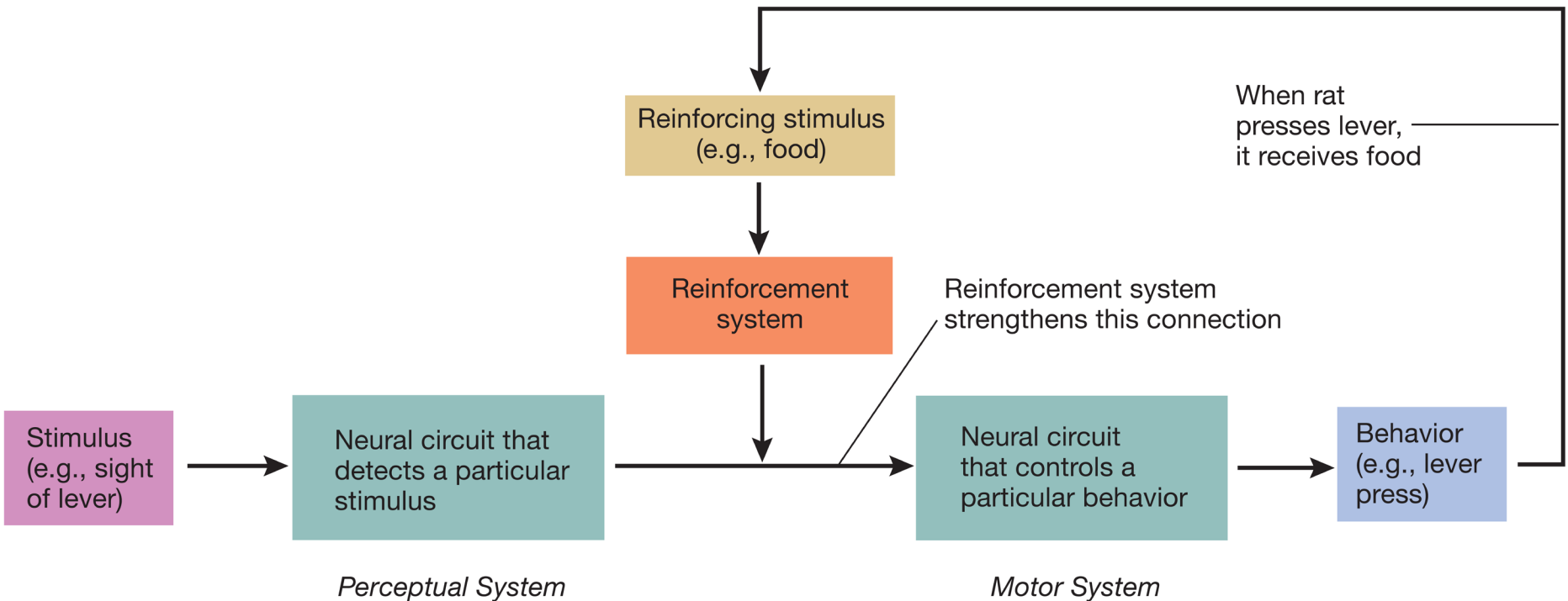


Withdrawal effects of heroin?



Figure 13.4

A Simple Neural Model of Operant Conditioning



Craving and Relapse



Figure 17.15

The Reinstatement Procedure: A Model of Relapse

The graph shows the acquisition of lever pressing for injections of a reinforcing drug during the self-administration phase and the extinction of lever pressing when the drug was no longer administered. A “free” shot of the drug or presentation of a cue associated with the drug during acquisition will reinstate responding. Source: Kalivas, P. W., Peters, J., & Knackstedt, L. (2006). Animal models and brain circuits in drug addiction. *Molecular interventions*, 6(6), 339.

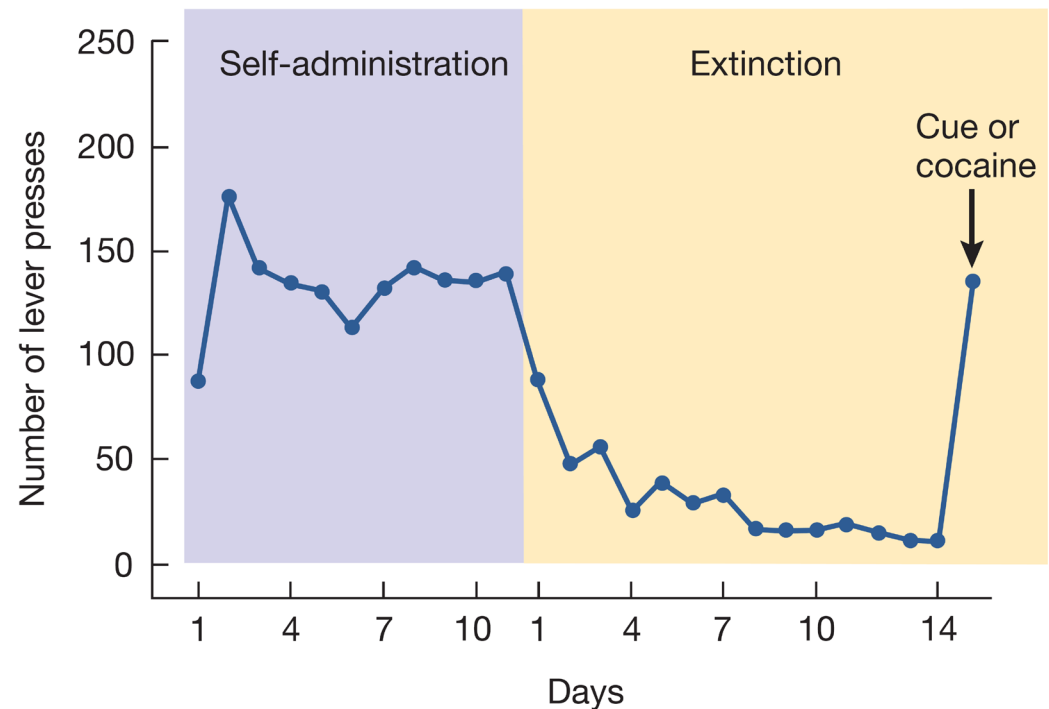
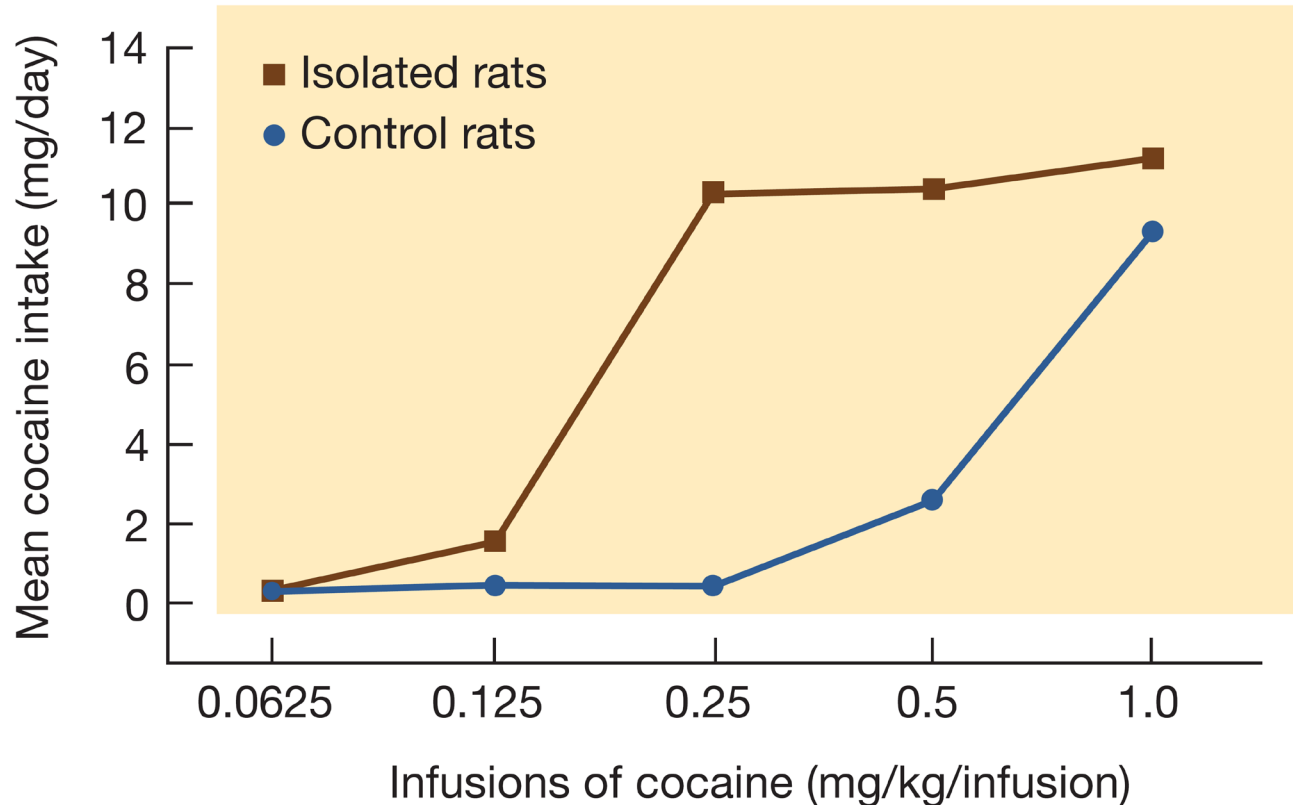


Figure 17.16

Social Stress and Cocaine Intake



The graph shows the cocaine intake of control rats and rats subjected to isolation stress early in life. Source: Kosten, T. A., Miserendino, M. J., & Kehoe, P. (2000).

Enhanced acquisition of cocaine self-administration in adult rats with neonatal isolation stress experience. *Brain research*, 875(1–2), 44–50.

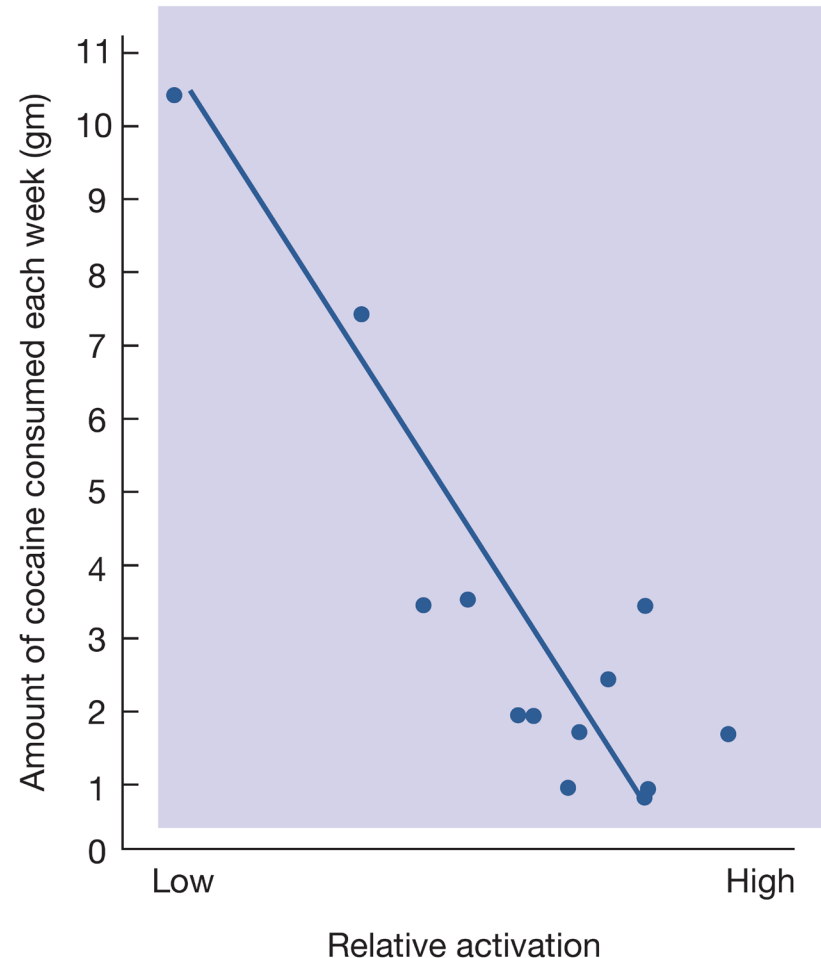
Craving and Relapse continued

- Less activation of the medial prefrontal cortex
 - Deficits on tasks and structural abnormalities
 - Involved in the extinction of conditioned emotional response
- High level of comorbidity of schizophrenia and substance use disorders
 - Up to half of those with schizophrenia have a substance abuse disorder
 - 70-90% of those with schizophrenia are nicotine dependent

Figure 17.17

Cocaine Intake and the Medial Prefrontal Cortex

The graph shows the relative activation of the medial prefrontal cortex as a function of the amount of cocaine normally taken each week by people who abused the drug. Source: Based on Bolla, K., Ernst, M., Kiehl, K., Mouratidis, M., Eldreth, D., Contoreggi, C., . . . & Funderburk, F. (2004). Prefrontal cortical dysfunction in abstinent cocaine abusers. *The Journal of neuropsychiatry and clinical neurosciences*, 16(4), 456–464.



Commonly Abused Drugs: Opiates

Opiates

- Opium derived from the opium poppy
- Many prescription pain relievers and designer drugs contain synthetic opiates
- Stimulate opiate receptors on neurons in the brain
 - Injections of opiates into both ends of the mesolimbic dopaminergic pathway are reinforcing

Newer types of opiates

- Fentanyl is a powerful synthetic opioid analgesic that is similar to morphine but is 50 to 100 times more potent.
- Fentanyl is known by such names as Actiq®, Duragesic®, and Sublimaze®. Street names for fentanyl or for fentanyl-laced heroin include Apache, China Girl, China White, Dance Fever, Friend, Goodfella, Jackpot, Murder 8, TNT, and Tango and Cash

Fentanyl

- People can swallow, snort, or inject fentanyl, or they can put blotter paper in their mouths so that fentanyl is absorbed through the mucous membrane.
- Fentanyl works by binding to the body's opioid receptors, which are found in areas of the brain that control pain and emotions
- Fentanyl's effects resemble those of heroin and include euphoria, drowsiness, nausea, confusion, constipation, sedation, tolerance, addiction, respiratory depression and arrest, unconsciousness, coma, and death.

Why death?

- Opioid receptors are also found in the areas of the brain that control breathing rate. High doses of opioids, especially potent opioids such as fentanyl, can cause breathing to stop completely, which can lead to death
- Fentanyl sold on the street can be mixed with heroin or cocaine, which markedly amplifies its potency and potential dangers
- “Know your pharmacist” quote by Dr. Patton

Three Waves of the Rise in Opioid Overdose Deaths

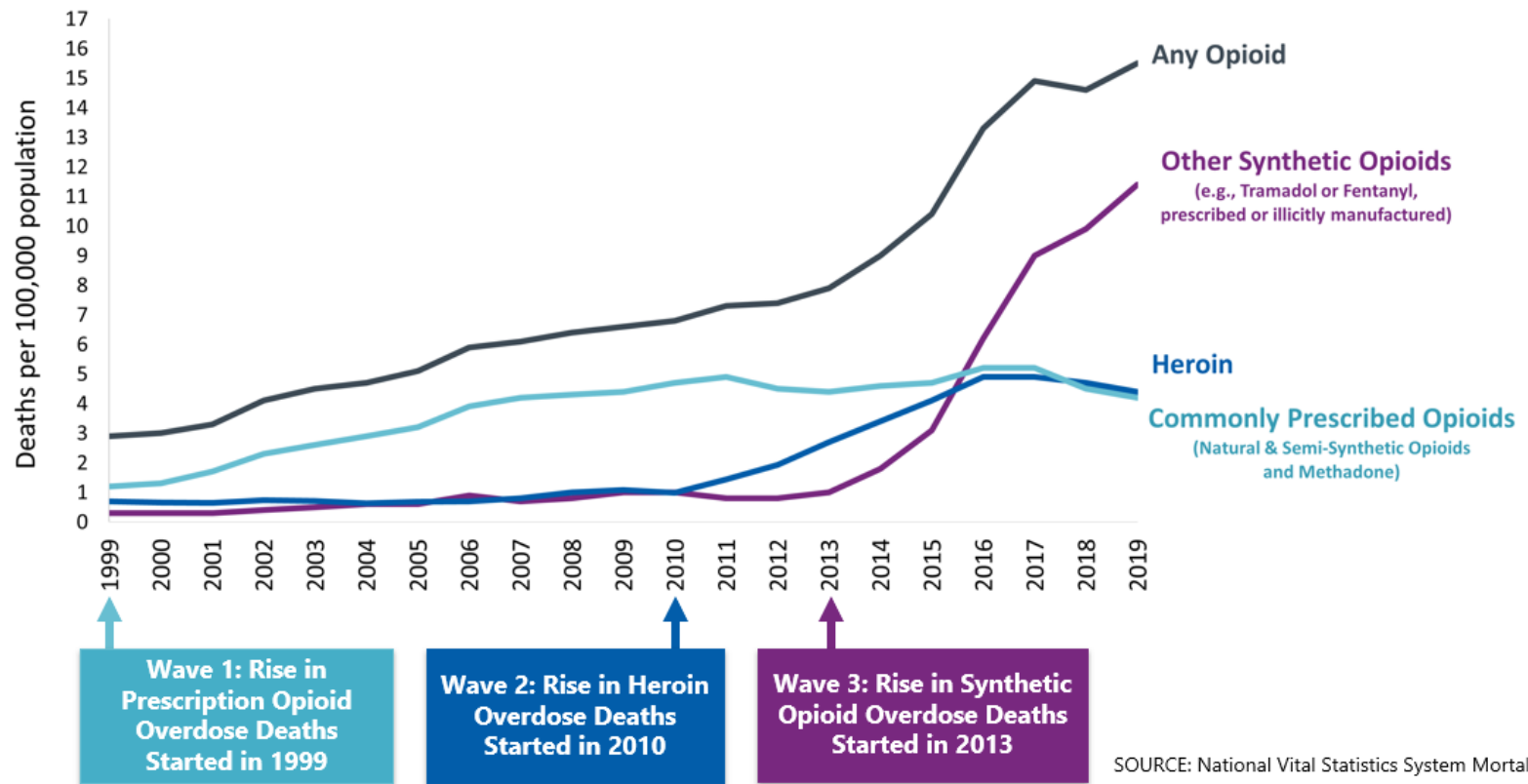
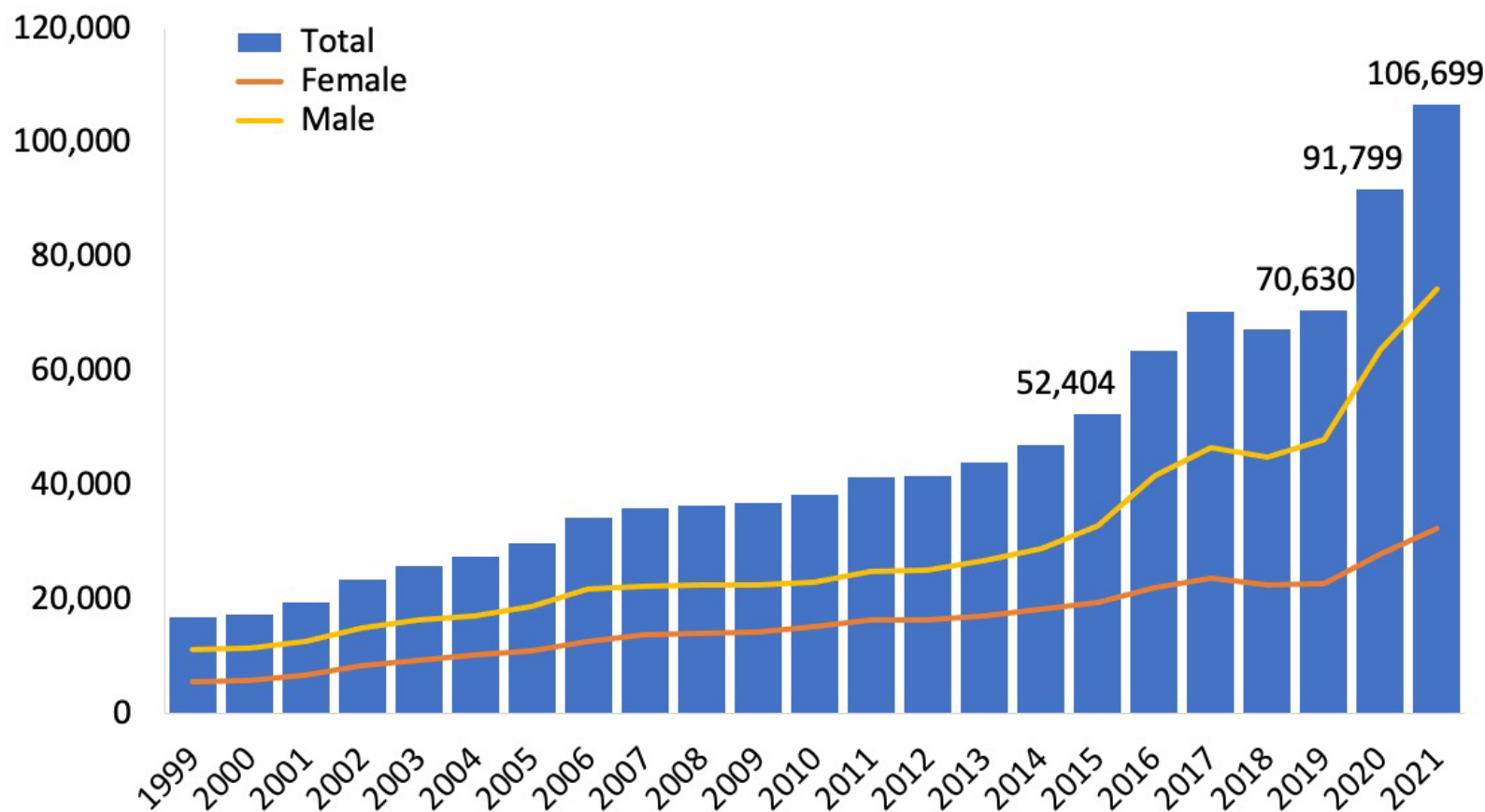
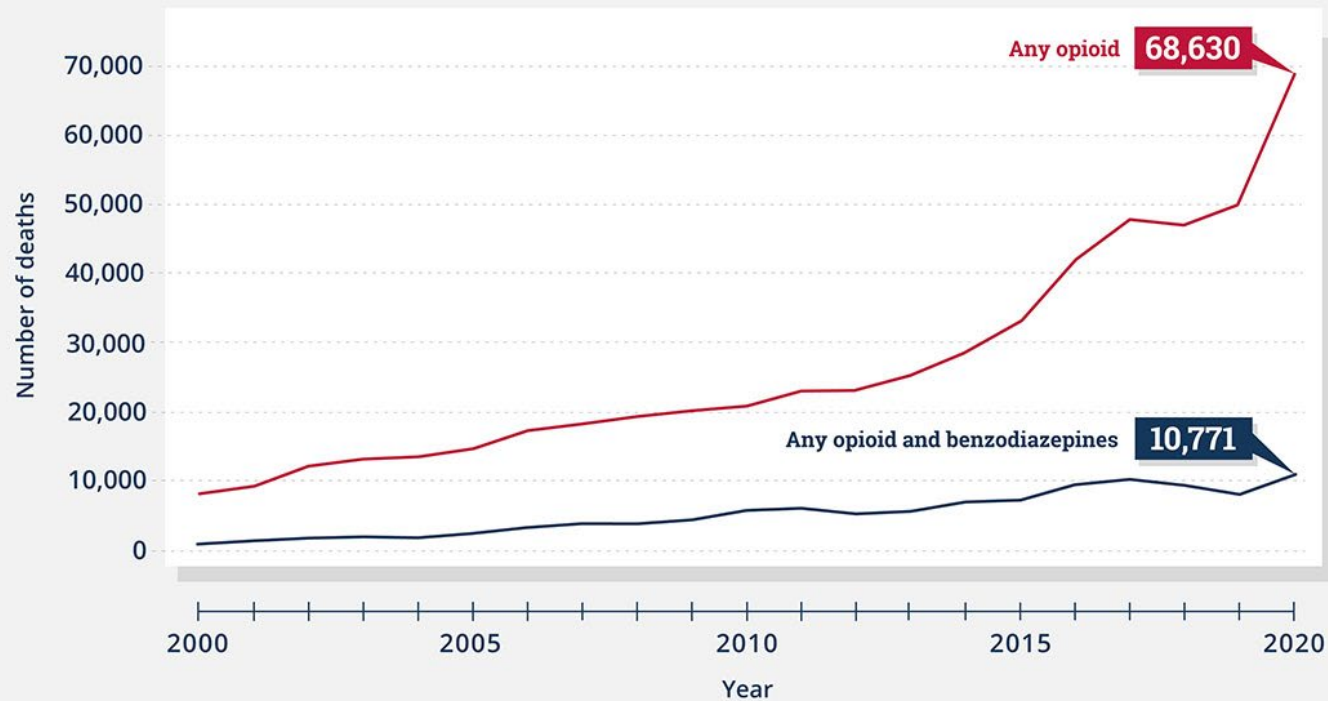


Figure 1. National Drug-Involved Overdose Deaths*, Number Among All Ages, by Gender, 1999-2021



*Includes deaths with underlying causes of unintentional drug poisoning (X40–X44), suicide drug poisoning (X60–X64), homicide drug poisoning (X85), or drug poisoning of undetermined intent (Y10–Y14), as coded in the International Classification of Diseases, 10th Revision. Source: Centers for Disease Control and Prevention, National Center for Health Statistics. Multiple Cause of Death 1999-2021 on CDC WONDER Online Database, released 1/2023.

National Overdose Deaths Involving Any Opioid, by Benzodiazepine* Involvement, All Ages, 2000 – 2020



*Among deaths with drug overdose as the underlying cause, the benzodiazepine category was determined by the T402.4 ICD-10 multiple cause of death code; the any opioid category was determined by the T40.0-T40.4, T40.6 ICD-10 codes.

Source: Centers for Disease Control and Prevention, National Center for Health Statistics. Multiple Cause of Death 2000–2020 on CDC WONDER Online Database, released 12/2021.

Newer types of opioids

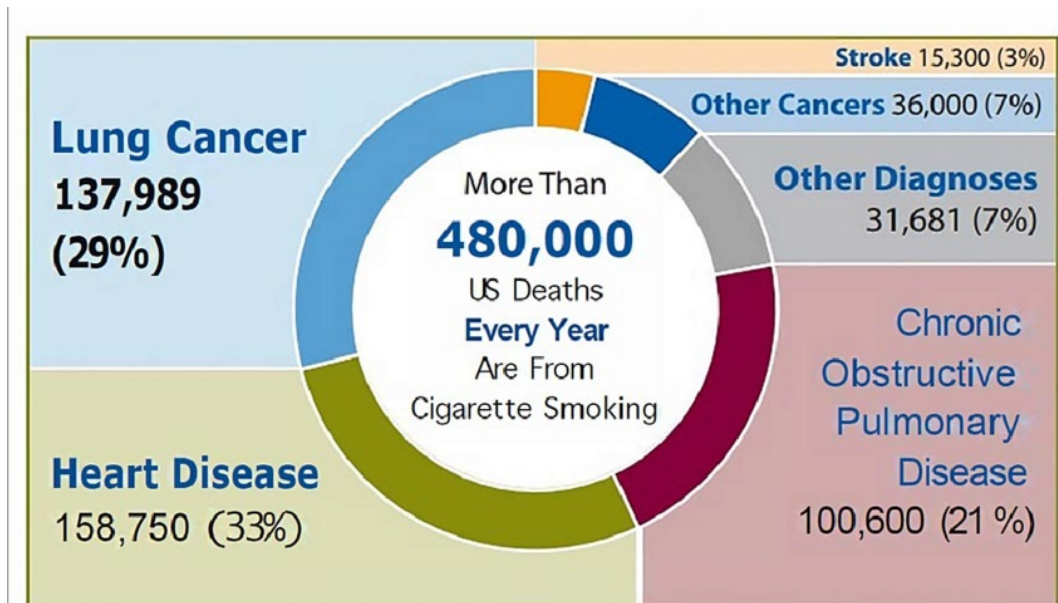
- W-18, the synthetic opioid was the most powerful in a series of about 30 compounds concocted at the University of Alberta and patented in the U.S. and Canada in 1984.
 - one experts say is 100 times more potent than fentanyl and 10,000 stronger than morphine

Commonly Abused Drugs: Stimulants

- Cocaine and amphetamine are potent dopamine agonists
 - Cocaine deactivates the dopamine transporter proteins
 - Amphetamine stimulates the release of dopamine
- Cocaine users are euphoric, active, and talkative
- Laboratory animals quickly learn to self-administer cocaine intravenously
 - They can self-inject so much that they die

Commonly Abused Drugs: Nicotine

- Accounts for more deaths than other drugs
- A very high abuse potential
- Stimulates nicotinic acetylcholine receptors and releases dopamine
 - The reinforcing effect is caused by activation of nicotinic receptors in the VTA



Note: Average annual number of deaths for adults aged 35 or older, 2005-2009.

Cigarettes and Death

Cigarette smoking causes about one of every five deaths in the United States each year. Cigarette smoking is estimated to cause the following:

- More than 480,000 deaths annually (including deaths from secondhand smoke)
- 278,544 deaths annually among men (including deaths from secondhand smoke)
- 201,773 deaths annually among women (including deaths from secondhand smoke)

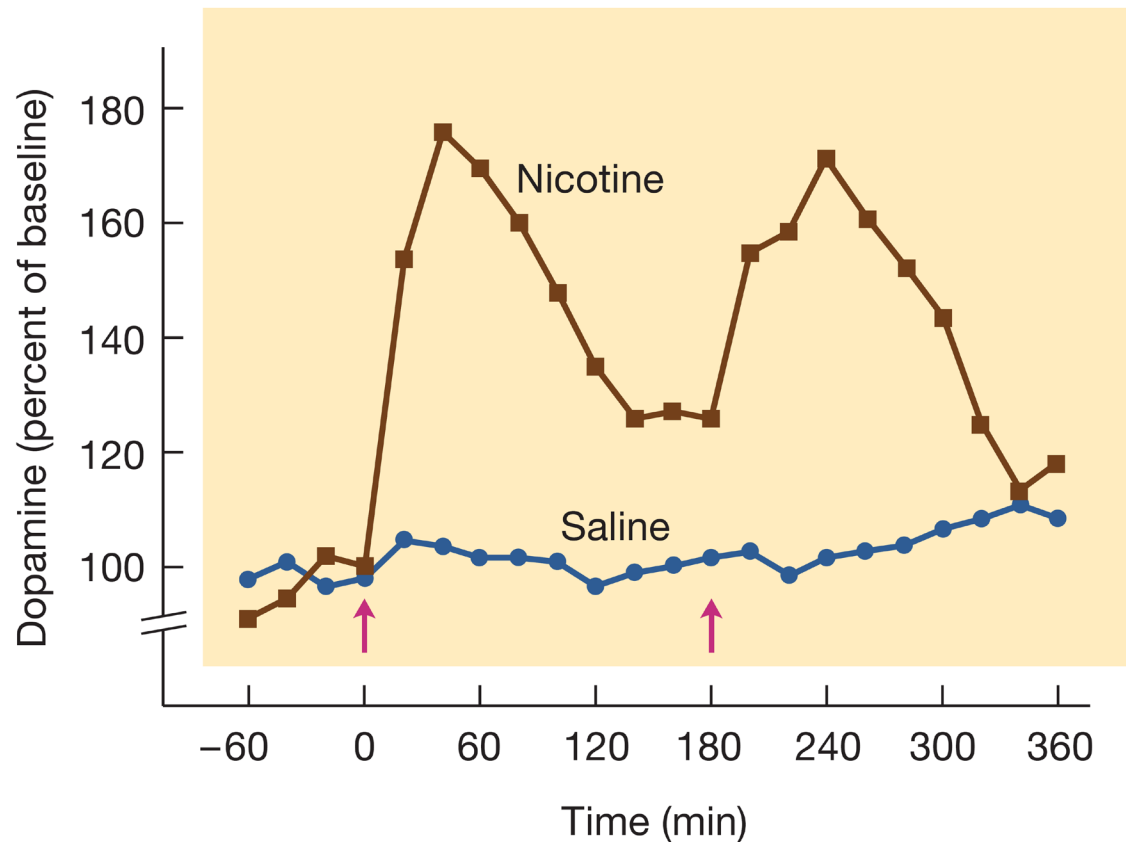
Cigarette smoking causes premature death:

- Life expectancy for smokers is at least 10 years shorter than for nonsmokers.
- Quitting smoking before the age of 40 reduces the risk of dying from smoking-related disease by about 90%.

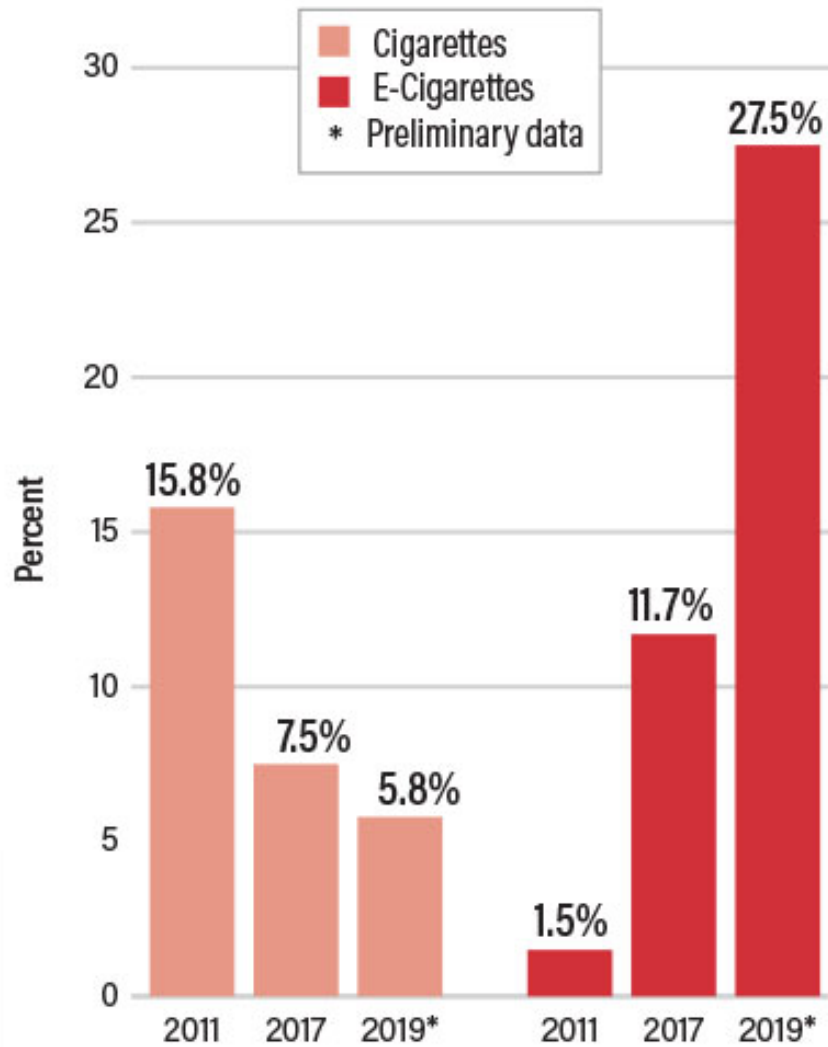
Figure 17.18

Nicotine and Dopamine Release in the Nucleus accumbens

The graph shows changes in dopamine concentration in the nucleus accumbens, measured by microdialysis, in response to injections of nicotine or saline. The arrows indicate the time of the injections. Source: Based on Damsma, G., Day, J., & Fibiger, H. C. (1989). Lack of tolerance to nicotine-induced dopamine release in the nucleus accumbens. *European Journal of Pharmacology*, 168(3), 363–368.



Youth Use of E-Cigarettes Far Surpasses That of Traditional Cigarettes



Source: National Youth Tobacco Survey

Commonly Abused Drugs: Alcohol

- Indirect antagonist at NMDA receptors and indirect agonist at GABA_A receptors
- Produces both positive and negative reinforcement
 - Release of dopamine in the NAC
 - The reinforcing effect is partly caused by its ability to trigger the release of endogenous opioids
- Effects of alcohol withdrawal are serious and can be fatal

NICOLAS CAGE

ELISABETH SHUE

LEAVING LAS VEGAS

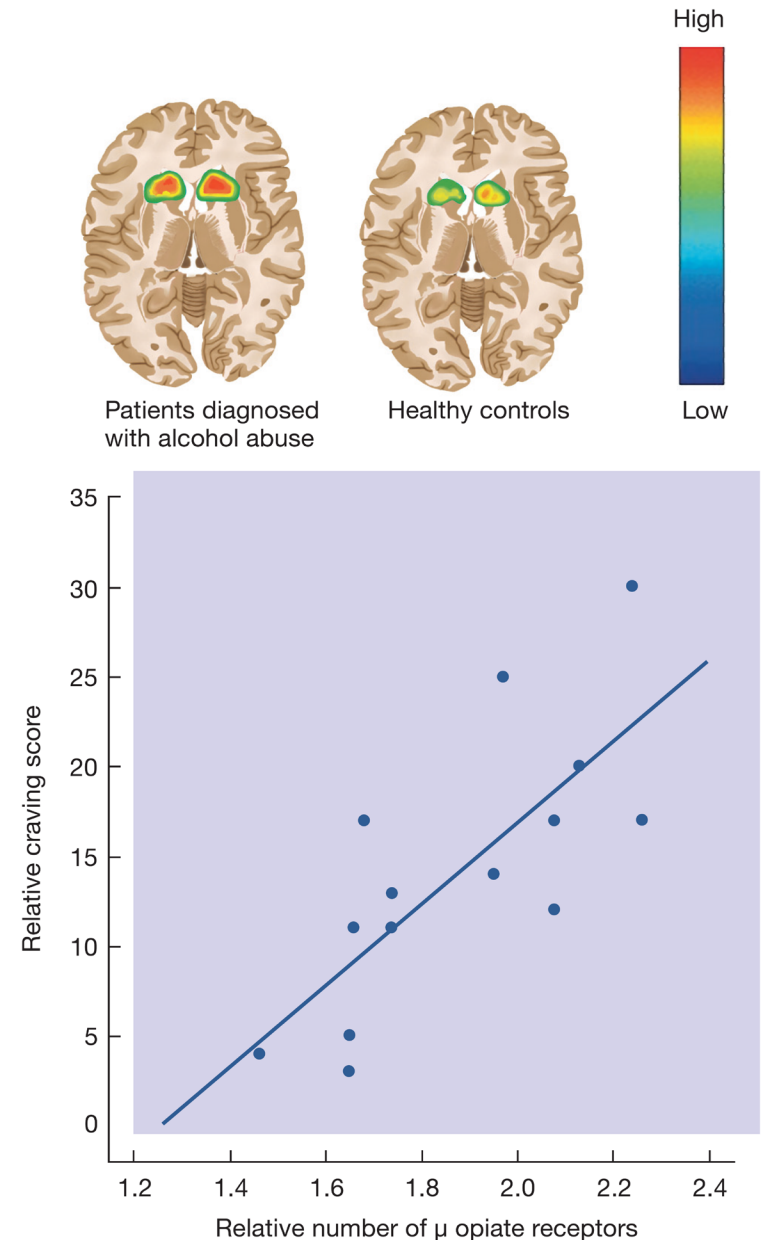


ORIGINAL MOTION PICTURE SOUNDTRACK

Figure 17.19

Cravings for Alcohol and μ Opiate Receptors

The drawings of the results of PET scans show the presence of μ opiate receptors in the dorsal striatum of detoxified patients diagnosed with alcohol abuse and healthy control participants. The graph shows the relative alcohol craving score as a function of relative numbers of μ opiate receptors. Source: Based on Heinz, A., Reimold, M., Wrase, J., Hermann, D., Croissant, B., Mundle, G., . . . & Bares, R. (2005). Correlation of stable elevations in striatal μ -opioid receptor availability in detoxified alcoholic patients with alcohol craving: A positron emission tomography study using carbon 11-labeled carfentanil. *Archives of General Psychiatry*, 62(1), 57–64.



Commonly Abused Drugs: Cannabis

- THC produced by the marijuana plant or synthetic cannabinoids
- Stimulates the CB1 receptor
- Produces memory impairment by disrupting the normal functions of the hippocampus
- Incidence of psychotic disorders is increased in cannabis users, especially long-term
- THC is a partial agonist of cannabinoid receptors and CBD is an antagonist



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March 19, 2025

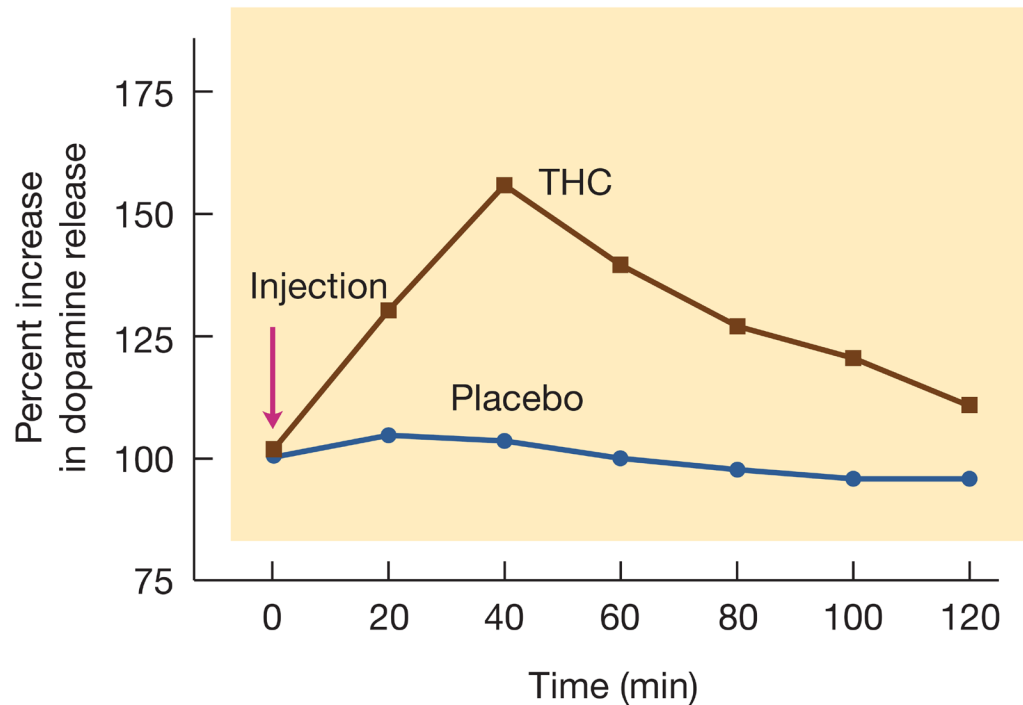
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SENATE APPROVES TOTAL THC BAN

(AUSTIN) — No product containing any amount of any form of THC, the psychoactive chemical present in cannabis, could be legally sold in Texas under a bill passed by the Senate on Wednesday. The bill comes in response to the explosion of consumable hemp products containing the drug, exploiting a loophole in a 2019 law intended to create an industrial hemp industry in Texas. That measure legalized hemp plants with less than a third of a percent concentrations of THC in a bid to foster a new non-consumable agricultural industry in Texas. The author of that bill, Lubbock Senator Charles Perry, said that instead, a synthetic THC market has arisen offering products significantly more intoxicating than ever before. "It's not your granddad's pot," he said. "It is something that is marketed as safe and legal — and it is anything but." Since 2019, Perry said that retailers have opened over 8,000 locations selling these products. He also authored Wednesday's bill, SB 3, which targets natural and synthetic forms of THC, but wouldn't affect the status of two other popular, non-

Figure 17.20

THC and Dopamine Secretion in the Nucleus Accumbens



The graph shows changes in dopamine concentration in the nucleus accumbens, measured by microdialysis, in response to injections of THC or placebo. Source: Chen, J., Paredes, W., Li, J., Smith, D., Lowinson, J., & Gardner, E. L. (1990). Δ 9-tetrahydrocannabinol produces naloxone-blockable enhancement of presynaptic basal dopamine efflux in nucleus accumbens of conscious, freely-moving rats as measured by intracerebral microdialysis. *Psychopharmacology*, 102(2), 156–162.

What is the ideal treatment for substance abuse?

Treatment for Substance Use Disorders

(1 of 5)

Therapeutic Interventions

- Must be supported by research as effective treatment
- Successful treatment is challenging
 - 40–60 percent are abstinent one year after a substance abuse intervention

Treatment for Substance Use Disorders: Opiates

- Methadone maintenance increases the opiate level in the brain slowly and does not produce a high
- Buprenorphine is a partial agonist for the μ opiate receptor
 - Blocks the effects of opiates and produces a weak opiate effect
 - No abuse potential

Treatment for Substance Use Disorders

(3 of 5)

Stimulants

- Vaccines that prevent cocaine molecules from crossing the blood–brain barrier
 - Become less sensitive to the activating effects of cocaine

Treatment for Substance Use Disorders

(4 of 5)

Nicotine

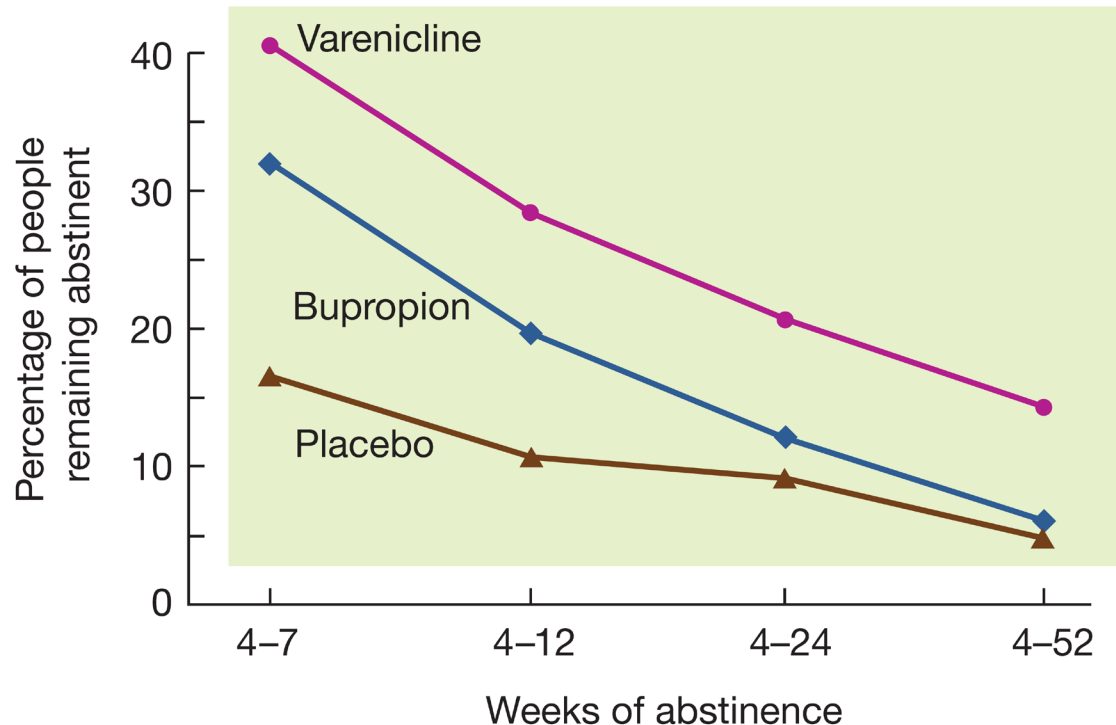
- Nicotine maintenance therapy using chewing gum or transdermal patches
- Bupropion is an antidepressant drug that serves as a catecholamine reuptake inhibitor
- Varenicline drug serves as a partial agonist for the nicotinic receptor
- Deep brain stimulation (DBS) and transcranial magnetic stimulation (TMS) may also help

Figure 17.21

Varenicline as a Treatment for Smoking

The graph shows the percentage of people who smoked that were treated with varenicline, bupropion, or placebo that abstained from cigarette smoking.

Source: Based on Nides, M., Oncken, C., Gonzales, D., Rennard, S., Watsky, E. J., Anziano, R., & Reeves, K. R. (2006). Smoking cessation with varenicline, a selective $\alpha 4 \beta 2$ nicotinic receptor partial agonist: results from a 7-week, randomized, placebo-and bupropion-controlled trial with 1-year follow-up. *Archives of Internal Medicine*, 166(15), 1561–1568.



Treatment for Substance Use Disorders

(5 of 5)

Alcohol

- Naltrexone is an opiate antagonist that decreases the reinforcing value of alcohol
- Acamprosate, an NMDA-receptor antagonist, reduces likelihood of drinking

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Neuropharmacology

Volume 40, Issue 6, May 2001, Pages 749-760



The anti-craving compound acamprosate acts as a weak NMDA-receptor antagonist, but modulates NMDA-receptor subunit expression similar to memantine and MK-801

G Rammes ¹  , B Mahal ^a, J Putzke ^b, C Parsons ^c, P Spielmanns ^c, E Pestel ^a, R Spanagel ^a, W Zieglgänsberger ^a, J Schadrack ^a

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Figure 17.22

Naltrexone as a Treatment for Alcoholism

The graphs show the proportion of patients who abstained from drinking while receiving naltrexone or a placebo. Source:

O'Brien, C. P., Volpicelli, L. A., & Volpicelli, J. R. (1996). Naltrexone in the treatment of alcoholism: a clinical review. *Alcohol*, 13(1), 35–39.

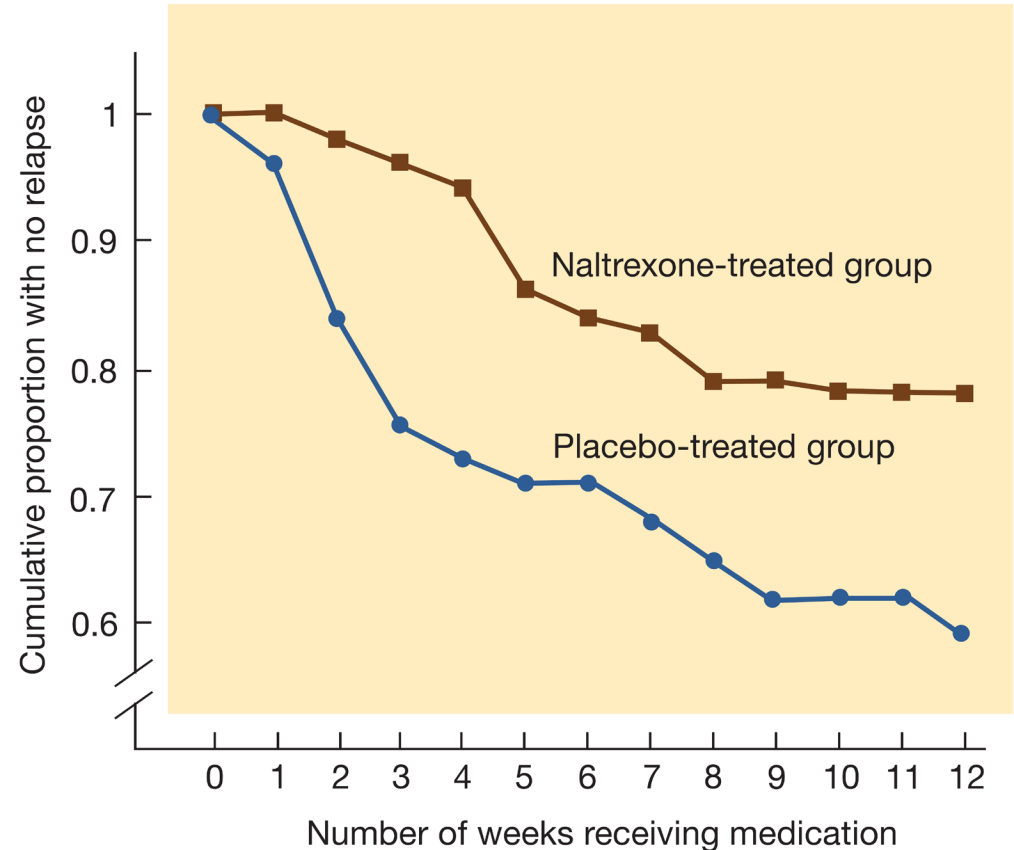
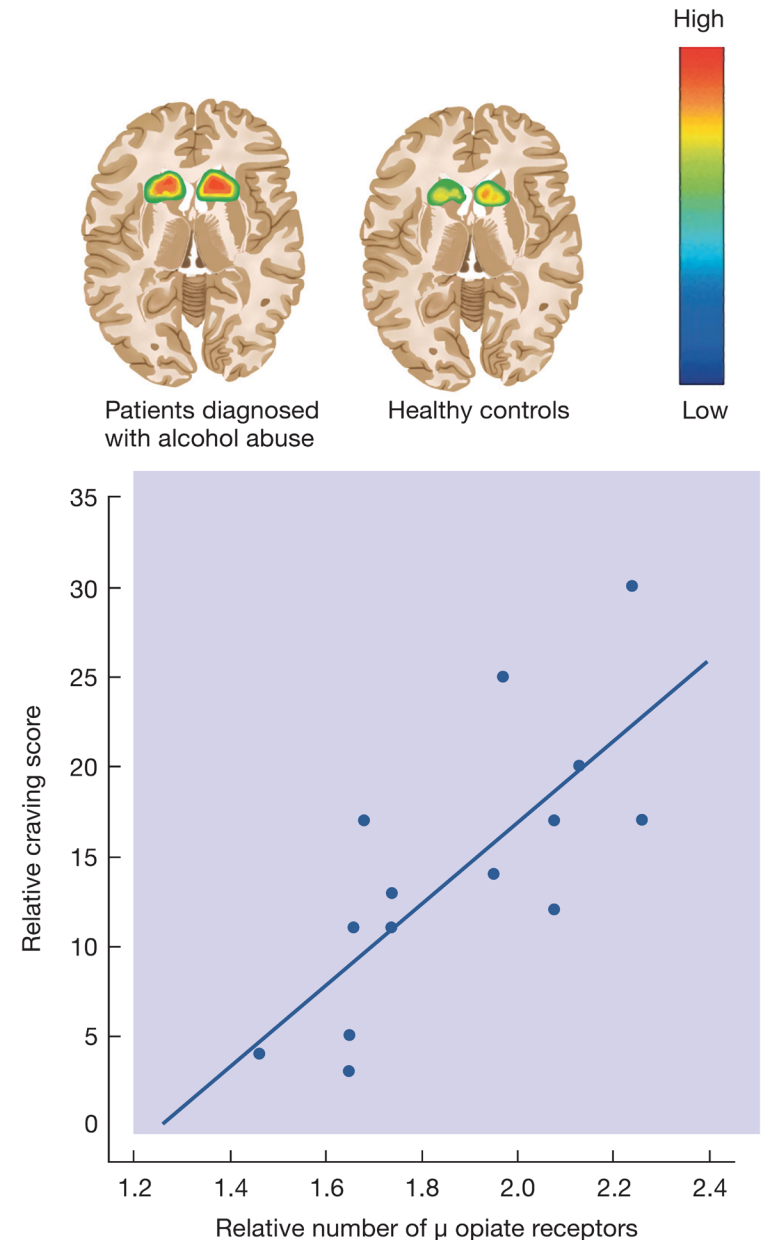


Figure 17.19

Cravings for Alcohol and μ Opiate Receptors

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What is Antabuse?

Antabuse is a brand name for disulfiram.

Disulfiram does not decrease alcohol cravings but blocks part of the metabolism process of alcohol, in turn, creating unpleasant feelings. This creates an aversion to the consequences of substance rather than stopping cravings.

About 5-10 minutes after ingesting alcohol, the patient begins to experience these symptoms for 30 minutes to an hour:



Accelerated
Heart rate



Flushed
Skin



Throbbing
Headache



Mental
Confusion



Visual
Disturbance

Symptoms emulate a severe hangover



Learning Objectives (1 of 3)

- **17.1** Describe the symptoms and possible causes of autism spectrum disorder.
- **17.2** Describe the symptoms and possible causes of ADHD.
- **17.3** Compare the SAM system and HPA axis in coordinating a stress response.
- **17.4** Describe the negative health outcomes associated with chronic stress.

Learning Objectives (2 of 3)

- **17.5** Compare the effects of long-term glucocorticoid exposure and early nurturing experiences on the brain in response to stress.
- **17.6** Summarize the relationship between stress and infectious disease.
- **17.7** List the symptoms of PTSD.
- **17.8** Describe the roles of genetic and environmental factors in the development of PTSD.

Learning Objectives (3 of 3)

- **17.9** Summarize treatments for PTSD.
- **17.10** Describe how positive and negative reinforcement contribute to the development of substance use disorders.
- **17.11** Describe commonly abused drugs.
- **17.12** Summarize the research on effective treatment for substance use disorders.